CPY 903

Advanced Assessment and Diagnosis

A Textbook in Counselling Psychology

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Advanced Assessment and Diagnosis

A Doctoral-Level Textbook in Counselling Psychology

[AUTHOR TO SUPPLY]

First edition

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Abbreviations

Abbreviations used in this text

Abbreviation	Expansion
AERA	American Educational Research Association
APA	American Psychological Association
ASD	Autism spectrum disorder
AUDIT	Alcohol Use Disorders Identification Test

PART I

Foundations and Measurement

The purposes of assessment, the logic of clinical inference, interviewing, and the psychometric basis of score interpretation.

Chapter 1 · Chapter 2 · Chapter 3 · Chapter 4

CHAPTER 1

Foundations of Advanced Psychological Assessment and Diagnosis

Learning Objectives

After working through this chapter, the reader should be able to:

- explain the purposes that psychological assessment serves in counselling psychology, and distinguish assessment from testing;
- describe assessment as a sequence of decisions in which each step constrains the next;
- translate a referral question into an assessment plan with proportionate methods and defensible limits;
- reason with base rates, competing hypotheses and known sources of error when interpreting findings;
- distinguish diagnosis, case formulation and treatment planning as related but non-identical products;
- identify the technical, cultural and ethical limits that qualify any assessment conclusion.

Key terms: assessment versus testing · referral question · competing hypotheses · case formulation · differential diagnosis · base rate · functional impairment · evidence-based assessment · multimethod assessment · decision threshold · interpretive limit

Introduction

Advanced assessment is not a longer version of routine testing. What changes at doctoral level is the *quality of the inference* the practitioner is

willing to defend. A student who can administer and score a battery may still be unable to say what the scores mean for this person, at this moment, in this setting, and with what margin of error. The purpose of this book is to develop that second competence: the disciplined movement from data to a conclusion that another qualified professional could inspect, challenge and, where necessary, overturn.

Three commitments organise the chapters that follow. The first is that assessment exists to answer a question that matters. Instruments, interviews and records are means; the referral question is the end that gives them sense. The second is that every conclusion is probabilistic. Scores carry error, informants disagree, and diagnostic categories are useful simplifications of heterogeneous presentations. The third is that assessment is an ethical act. Its conclusions can open or close access to treatment, education, employment, custody and liberty, and the assessor therefore owes the client a transparent account of what was measured, what was inferred and what remains uncertain.

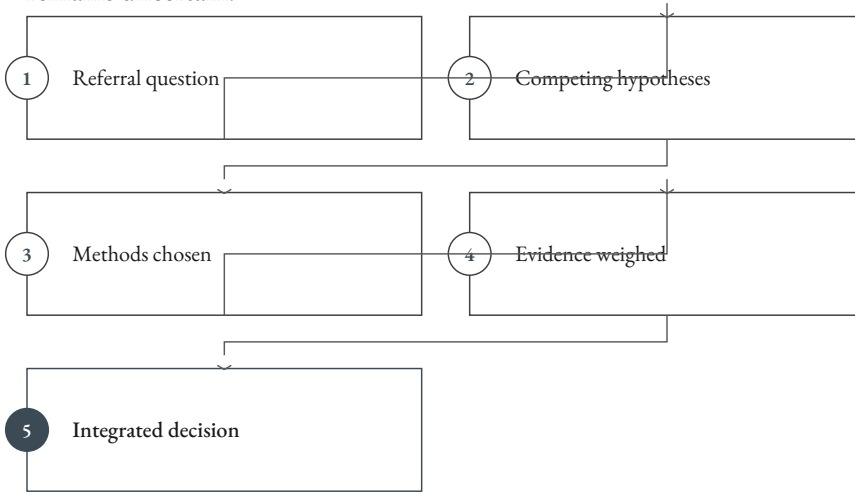


Figure 1.1. The movement from referral question to responsible decision. Each step constrains the next; the final decision inherits the uncertainties recorded at every earlier step.

Figure 1.1 presents the logic of the book in compressed form. The remaining sections of this chapter elaborate each element of that sequence, and the chapters that follow apply it to interviewing, measurement, specific

instrument families and specific clinical presentations. Throughout, the emphasis falls on reasons rather than labels: a diagnosis that cannot be justified in writing is not yet a diagnosis, only an impression.

1.1 The Place of Assessment in Counselling Psychology

Counselling psychology has historically defined itself by attention to the person in context, to developmental and relational history, and to strengths as well as deficits. Assessment in this tradition is therefore not a diagnostic ritual performed before therapy begins, but a continuous and collaborative activity that informs formulation, monitors change and structures the ending of treatment. Its distinctive contribution is the integration of standardised measurement with an account of meaning: what the difficulty is, what it means to this person, and what function it serves in their life.

Assessment and testing are not synonyms

Testing is a procedure: a standardised stimulus set, a specified response format, a scoring rule and a norm or criterion against which performance is compared. Assessment is the broader process of which testing may or may not form a part. It begins with a question, selects methods proportionate to that question, integrates data from multiple sources, and ends with a communication that is useful to somebody. Meyer and colleagues (2001) demonstrated that psychological tests generally achieve respectable validity for the constructs they measure, but the same evidence shows that validity attaches to *interpretations* rather than to instruments. A test does not carry its meaning with it; the assessor supplies the meaning and is accountable for it.

This distinction has practical consequences. It explains why a technically well-administered battery can still yield a poor assessment, and why an assessment conducted without standardised instruments can, in some circumstances, be defensible. It also explains why the competence required for advanced practice cannot be reduced to test administration: the difficult skills are question formulation, method selection, integration

and communication.

The institutional context of assessment

Assessment is commissioned by someone. Referrals arrive from general practitioners, psychiatrists, schools, courts, employers, insurers, adoption agencies and, increasingly, from clients themselves. Each source brings an implicit theory of what the problem is and what a useful answer would look like. A paediatric referral asking whether a child "has ADHD" has already framed the difficulty as an individual disorder rather than, say, a classroom-fit problem; a court asking about parenting capacity has framed it as a prediction about future behaviour. Part of advanced competence is recognising these frames early, negotiating them where they are inadequate, and recording the negotiation.

Counselling psychologists also work in settings where the assessment itself is an intervention. A careful history can be the first occasion on which a client's experience has been taken seriously and organised coherently; feedback can reframe a self-condemnatory narrative. Finn and Tonsager (1997) argued that assessment information can produce measurable therapeutic benefit when it is shared collaboratively, and this finding has been absorbed into what is now called therapeutic or collaborative assessment. The implication is not that assessment must always be therapeutic, but that its interpersonal dimension is never neutral.

Assessment as a professional identity

For the doctoral practitioner, assessment competence is a marker of professional standing. It signals the capacity to hold technical and relational knowledge together: to know the standard error of a difference, and also to know that a client who arrives twenty minutes late may be communicating something about the assessment. The chapters of this book are organised to develop both capacities simultaneously, because in practice they cannot be separated.

1.2 Assessment as a Decision Process

An assessment is best understood as a decision tree rather than a data-collection exercise. The first decision concerns scope: which questions are answerable within the resources available, and which must be referred elsewhere. The second concerns methods. The third concerns thresholds: how much evidence is required before a conclusion is offered. The fourth concerns communication: what the client, the referrer and any other legitimate audience need to know. Each decision narrows the space of possible conclusions, and each should be revisable in the light of new evidence.

Sequential and iterative structure

Although assessment plans are written in sequence, real assessments loop. A discrepancy between interview and questionnaire may send the assessor back to the history; a collateral report may reopen a question that had appeared settled. Explicitly acknowledging this iterative structure protects against the common error of treating the initial plan as a contract. In practice, three to five significant revisions of a hypothesis set are normal in a complex adult assessment, and records that show no revision at all usually indicate that no genuine hypothesis was ever tested.

Efficiency is an ethical requirement

Because assessment consumes the client's time, money and emotional resources, proportionate selection of methods is an ethical as well as an economic matter. Every additional instrument must be justified by the probability that it will change a decision. This principle, sometimes called incremental validity, is the strongest available argument against the indiscriminate batteries that characterise inexperienced practice. Where a battery is retained for reasons of convention or institutional requirement, the assessor should say so rather than implying that each instrument earned its place.

Documenting decisions

The decision record is the assessor's protection and the client's guarantee. A workable record states the referral question in the referrer's own terms, the questions the assessor intended to answer, the methods chosen and the reasons for them, the hypotheses under consideration, the evidence that would discriminate between them, and the point at which the evidence was judged sufficient. This record need not be long, but it must be specific enough that a colleague could reconstruct the reasoning and identify precisely where they would have decided differently.

1.3 From Referral Question to Assessment Plan

Referral questions vary in precision from the unusable ("please assess") to the over-specified ("please confirm PTSD"). The assessor's task is to convert whatever arrives into one or more answerable questions, each with a stated decision attached. "Is this person depressed?" is not answerable in the abstract; "Is the current presentation better explained by a depressive episode than by hypothyroidism, grief or burnout, and would the answer change the next treatment step?" is.

Qualities of a well-formed assessment question

A well-formed question specifies the population or person, the constructs at issue, the comparison or decision that the assessment must inform, and the standard of evidence the decision requires. It also specifies a time frame: whether the question concerns current state, recent episode or lifelong pattern determines whether state measures, retrospective interviewing or informant history will carry most weight. Where the referrer cannot supply these elements, the assessor supplies provisional ones and records them as assumptions.

Proportionality

The plan should be proportionate in three senses: in cost and burden relative to the stakes; in intrusiveness relative to the question; and in precision relative to the decision threshold. A screening question about alcohol use in a university counselling service may be answered by a

validated self-report questionnaire; a question about fitness to stand trial requires a different order of evidence, different qualifications and different reporting conventions. Proportionate planning also means anticipating what will *not* be done and stating why.

Planning for what could go wrong

Competent plans include contingencies. If the client does not attend, if an interpreter is needed, if records are unavailable, if the client's reading level prevents a self-report measure, if risk emerges mid-assessment — each contingency should have a planned response. Table 1.1 sets out the planning decisions that most often determine whether an assessment succeeds, together with the failure mode that each decision prevents.

Table 1.1. Planning decisions, their rationale, and the failure they prevent

Planning decision	Why it matters	Failure prevented
Define the decision the assessment must inform	**Methods are selected for their bearing on a decision, not for their prestige**	**Over-testing; reports that answer no question**
Specify constructs before selecting instruments	**Construct clarity prevents measuring convenient proxies**	**Score-led assessment in which the battery defines the problem**
State the required standard of evidence	**High-stakes decisions need converging, independent evidence**	**Conclusions that outrun the data**
Plan the comparison or alternative explanations	**Differential reasoning requires rival hypotheses to be explicit**	**Confirmation bias; premature closure**
Plan for language, literacy and sensory access	**Validity depends on comprehension of the task**	**Misinterpreted scores presented as ability**
Set the communication plan at the outset	**Feedback shapes whether findings are used**	**Reports that are filed but never acted upon**
Record what will not be assessed	Scope limits are part of the conclusion	Implied claims beyond competence or remit

1.4 Diagnosis, Formulation and Treatment Planning

Three products emerge from clinical assessment, and confusing them is one of the most common sources of poor practice. A *diagnosis* is a classification decision: does this presentation meet the criteria for a recognised disorder, and which one? A *formulation* is an explanatory account: why this person, at this time, in this context, has developed and maintained these difficulties? A *treatment plan* is a set of actions derived from both. Diagnostic classification is retrospective, rule-governed and comparatively reliable; formulation is prospective, theory-laden and comparatively personal.

What diagnosis does and does not do

Diagnosis does several things well. It supports communication between professionals, links presentation to an evidence base, underwrites access to services and reimbursement, and imposes a discipline of explicit criteria that reduces idiosyncratic labelling. It does not explain aetiology, predict individual response to treatment reliably, or capture the person's own understanding. Nor is it stable across all presentations: mild and mixed presentations frequently fail to meet full criteria while still requiring treatment, which is why residual and unspecified categories exist and why their use should be documented rather than apologised for.

Formulation as the bridge

Formulation uses assessment evidence to construct a testable account of onset, maintenance and change. A useful formulation specifies predisposing factors, precipitants, maintaining processes, protective resources and the practical barriers that limit recovery. It should be expressed in language the client can contest, and it should predict something: if the formulation is right, then a specified intervention should produce a specified change. Where a formulation makes no predictions, it is usually a description rather than an explanation.

From formulation to plan

Treatment planning translates formulation into sequenced action, and the translation is where assessment earns its practical value. A plan that names a therapy but not the target mechanism is a plan that cannot be evaluated. Good plans specify what will be addressed first, why that target was prioritised, what would count as progress, and what would prompt a return to assessment. They also acknowledge constraints: waiting lists, cost, family obligations and the client's own preferences legitimately shape what is feasible, and a plan that ignores them is a document about an imaginary client.

1.5 Dimensional and Categorical Thinking

Psychopathology can be described categorically, as the presence or absence of a disorder, or dimensionally, as a position on one or more continuous distributions. The two descriptions answer different questions and are not in competition. Categorical description is indispensable for communication, service eligibility and legal decision-making; dimensional description preserves information about severity and subthreshold variation that categories discard.

Evidence bearing on the distinction

Structural research on the classification of mental disorders has consistently found that comorbidity is the rule rather than the exception, that symptom distributions in the population are continuous rather than bimodal, and that the boundaries between neighbouring categories are porous. The current editions of both major classifications have responded unevenly to this evidence. The *Diagnostic and Statistical Manual of Mental Disorders*, fifth edition, text revision (DSM-5-TR; American Psychiatric Association, 2022) retains a predominantly categorical architecture while adding dimensional severity ratings, cross-cutting symptom measures and a hybrid model for personality disorders in its emerging-measures section. The eleventh revision of the *International Classification of Diseases* (ICD-11) characterises personality disorder by severity of disturbance with optional

trait qualifiers, a substantially dimensional departure (World Health Organization, 2024).

Consequences for the assessor

Dimensionally-minded assessors report severity and threshold information alongside any diagnosis: not only that criteria are met, but how far above threshold the presentation lies, and which criteria are marginal. They record subthreshold presentations explicitly rather than forcing them into residual categories. They treat a diagnostic boundary as a decision rule with an error rate rather than a fact about the person. And they resist the temptation to resolve genuine diagnostic uncertainty by choosing the category that seems least objectionable, a practice that is common, well-intentioned and usually misleading.

1.6 Sources of Assessment Evidence

Different methods answer different questions, and the most persistent error in advanced practice is asking one method to do the work of another. An interview yields meaning, chronology and the client's own account; systematic observation yields behaviour in context; informant report yields the perspective of someone who sees the client in settings the assessor never will; records yield documented history and prior reasoning; standardised measures yield comparison with a reference group. Table 1.2 summarises the distinctive contribution and characteristic limitation of each source.

Table 1.2. Sources of assessment evidence: contribution and characteristic limitation

Source	Principal contribution	Characteristic limitation
Clinical interview	**Meaning, chronology, context, the client's account**	**Vulnerable to recall bias, impression management and interviewer effects**
Behavioural observation	**Actual behaviour in a specific setting**	**Reactivity; limited sampling across settings**
Informant report	**Perspective of family, partners, teachers or carers**	**Informant distress, motive and opportunity to observe**
Records and prior reports	**Documented history; earlier reasoning and its errors**	**Uneven quality; outdated conclusions recycled uncritically**
Standardised tests	**Normative comparison; quantified error**	**Construct and population specificity; administration constraints**
Performance validity measures	**Whether effort and engagement were sufficient**	**Imperfect sensitivity; not a test of honesty**
Biological and medical investigations	**Ruling out or identifying medical contributors**	**Not diagnostic of psychological disorders; access and cost**
Self-monitoring and ecological measures	Behaviour in natural settings over time	Adherence; reactivity; data burden

Convergence and divergence

Multimethod assessment is valuable not because agreement is reassuring, though it is, but because disagreement is informative. A client who reports severe depressive symptoms while presenting as organised, socially engaged and functionally intact has supplied the assessor with a discrepancy that requires explanation. Possible explanations include situational fluctuation, concealment, insight differences, measurement reactivity, informant perspective or a genuine mismatch between symptom report and functional capacity. Each implies a different next step. Averaging the two sources destroys exactly the information that would have resolved the question.

Choosing among sources

Source selection follows from the question. Questions about frequency and topography of behaviour favour observation and self-monitoring; questions about developmental history favour records and informants; questions about current severity favour standardised state measures; questions about meaning favour interview. When resources permit only one source, the assessor should state in the report which claims rest on that source alone, and how confident the conclusion can therefore be.

1.7 Base Rates and Clinical Probability

One of the most consequential findings in the assessment literature is that clinicians reason poorly about probability. The base rate of a condition in the setting where the assessment occurs determines how much weight a positive finding can bear. Where a condition is rare, even a highly specific test generates more false positives than true positives, because the pool of people without the condition is so much larger. This is not a technicality; it is the arithmetic that governs screening programmes, forensic opinions and diagnostic impressions formed in specialist clinics.

The arithmetic in practice

Suppose a screening instrument with sensitivity of .90 and specificity of .90 is used in a setting where the condition affects 2 per cent of those assessed. Of 1,000 people, 20 have the condition and 18 are correctly identified; 980 do not, and 98 are incorrectly identified as positive. A positive result therefore carries a probability of roughly 16 per cent that the condition is present. In a setting where prevalence is 30 per cent, the same instrument yields a positive predictive value above 70 per cent. The instrument has not changed; the setting has. Interpreting a score without knowing the setting is therefore an arithmetic error, not merely a stylistic one.

Consequences for threshold setting

Base rates also govern where a decision threshold should sit. Where the cost of a missed case is high and the cost of a false positive is moderate —

suicidal risk in an emergency department, for instance — a lower threshold is defensible, provided the assessment plan includes a way to resolve the false positives. Where a positive result triggers an irreversible consequence — a child protection removal, a fitness-for-duty determination — the threshold must be higher and the evidence converging. The assessor's responsibility is to make this trade-off explicit rather than to pretend that a cut score carries intrinsic authority.

Clinician behaviour

Research on clinical judgement has documented systematic departures from normative probability reasoning, including insensitivity to base rates, overconfidence in sparse evidence and premature closure on an early hypothesis. The remedy is structural rather than exhortatory: routine statement of the setting's base rate, routine statement of the likelihood ratio implied by each finding, and routine review by a colleague before high-stakes conclusions are issued. These habits are more reliable than the intention to be unbiased, which everyone possesses and which almost no one can sustain unaided.

1.8 Functional Impairment and Context

Diagnosis describes the pattern of symptoms; impairment describes what the pattern prevents. The two are correlated but far from identical, and the difference matters for treatment planning, for eligibility decisions and for the ethical requirement not to pathologise understandable reactions. A person may meet criteria for a depressive episode while continuing to work effectively; another may fall well short of criteria while unable to leave the house. Assessment that reports only symptom counts has measured half of what the referrer needs.

Domains of functioning

Functioning is best assessed domain by domain rather than as a global impression. Work or study, household management, financial organisation, self-care, close relationships, parenting, social participation

and health behaviour each carry different demands and different consequences of failure. Within each domain, the assessor distinguishes capacity from performance: what the person could do under optimal conditions, and what they actually do in their current environment. The gap between the two is often the most clinically informative finding in the assessment.

Context as an explanatory resource

Context is not noise to be controlled but information to be interpreted. Behaviour that appears disordered in one setting may be adaptive in another: hypervigilance in a context of genuine danger, withdrawal in a context of relentless humiliation, perfectionism in a context of conditional approval. An assessor who records only the behaviour has described a symptom; an assessor who records the behaviour and the conditions that elicit and maintain it has described a person. This is the sense in which counselling psychology's attention to context is a technical asset rather than a sentimental preference.

Reporting impairment

Impairment should be reported in behavioural terms that the referrer can use: not "moderate impairment" alone, but what the person can no longer do, for how long, with what accommodation and at what cost. Where impairment is disputed — as it often is in disability, insurance and educational determinations — the assessor should describe the evidence on which the judgement rests, including the client's own account, observed performance, informant report and any documentary corroboration, and should state what was not verified.

1.9 Evidence-Based Assessment

Evidence-based practice in psychology requires the integration of the best available research evidence, clinical expertise and the client's characteristics, culture and preferences. Applied to assessment, the principle implies that instrument selection, interpretation and decision thresholds should each be

justified by evidence rather than convention. An instrument earns its place by demonstrated reliability and validity for the population and purpose at hand, by incremental validity over cheaper alternatives, and by evidence that its use improves outcomes.

What the evidence base covers

The evidence relevant to assessment is broader than test manuals. It includes research on the diagnostic accuracy of instruments and interviews, on clinical judgement and its biases, on the effects of feedback on treatment engagement, and on the outcomes of assessment-driven decision-making. Assessment research has been unevenly translated into routine practice: many widely used instruments have limited independent validation in the populations where they are most often applied, and some well-validated procedures are under-used because they are unfamiliar or laborious. The assessor's obligation is to know the state of the evidence for what they use, including its gaps.

Expertise and its limits

Clinical expertise is a legitimate component of evidence-based practice, but it is not a licence for idiosyncrasy. Expertise shows itself in the quality of question formulation, in the recognition of when a standard procedure does not fit, in the calibration of confidence to evidence, and in the willingness to seek consultation. It does not show itself in the confident impression that contradicts the data. Where expertise and evidence conflict, the conflict should be recorded and, where possible, tested rather than resolved by seniority.

Client characteristics and preferences

The third element of evidence-based practice is the client. Language, culture, age, education, sensory and cognitive capacity and the client's own goals all affect whether an assessment is valid and whether its conclusions are useful. A measure validated in one population and administered to a client from another is not automatically invalid, but its validity is an empirical question that the assessor must address rather than assume.

Chapter 14 treats this question in detail; here it is enough to note that cultural validity is a property of an interpretation, not a courtesy extended to a client.

1.10 Interprofessional Assessment

Complex presentations are rarely assessed by one profession alone. Psychiatrists contribute medical formulation and pharmacological reasoning; neurologists and physicians contribute differential medical explanation; speech and language therapists, occupational therapists and educational psychologists contribute functional and developmental expertise; social workers contribute safeguarding and systemic knowledge; nurses contribute observation across time. The counselling psychologist's contribution is typically the integration of psychological measurement with formulation, and the quality of that contribution depends on the quality of the exchange around it.

Communication across professional boundaries

Interprofessional assessment fails most often at the boundary rather than within each profession. Referrals arrive without the information needed to plan; reports are written in vocabulary the receiving professional cannot use; recommendations assume resources that do not exist in the receiving setting. Effective practice includes agreeing at the outset what each party will contribute, what format the output should take, and who holds responsibility for what. Where disagreements persist — about diagnosis, about risk, about capacity — the disagreement itself should be documented, because the next clinician will need to know that the question is contested.

Role boundaries and competence

Interprofessional working also requires clarity about competence. The assessment of neurological disorder, of medication effects, of legal capacity and of child protection risk each carry requirements that a doctoral qualification in counselling psychology does not by itself satisfy.

Recognising the boundary of one's competence, referring appropriately and contributing what one is qualified to contribute are marks of seniority rather than admissions of inadequacy.

1.11 Limits of Assessment Conclusions

Every assessment conclusion is bounded by the methods that produced it, the population in which those methods were validated, the conditions under which they were administered and the time at which they were made. Advanced practice consists partly in stating those bounds precisely. "Within the limits of a single-session assessment using self-report measures in a client whose literacy in the test language was uncertain, the evidence is consistent with a moderate depressive episode; alcohol use was not assessed and may affect the picture" is a conclusion that can be acted upon. The same finding reported as "moderate depression" cannot.

Sources of uncertainty

Uncertainty enters at four points. Measurement error affects every score and should be expressed as a confidence interval rather than a point estimate. Sampling error affects every inference from a normative group to an individual. Method variance means that different methods measure partly different things. And genuine indeterminacy exists where the available evidence is compatible with more than one explanation. Only the last is usually acknowledged in reports; the first three are more common and more tractable.

Time and change

A single assessment describes a period, not a person. Symptoms fluctuate with sleep, substances, medication, menstrual cycle, seasonal change, relationships and circumstance. Conclusions about stable traits require evidence of stability, and conclusions about change require repeated measurement with instruments capable of detecting it. Where an assessment is used to make a decision that will persist — a diagnosis that will follow a child through school, a determination of capacity — the

report should state its own expiry conditions: what change would require reassessment.

The limits of the assessor

Finally, the assessor's own limits belong in the account. Fatigue, unfamiliarity with a language or culture, a strong reaction to the client, an institutional pressure toward a particular conclusion: these are not disqualifying, but they are relevant. Consultation, supervision and, where appropriate, disclosure to the client are the available remedies. A report that records "assessment conducted without access to school records; conclusions about developmental history are provisional" is more professional than one that conceals the gap.

1.12 Integrating Findings into Responsible Decisions

Integration is the point at which the book's central skill is exercised. It requires holding several kinds of evidence in view at once, weighing them according to their quality and relevance, and producing a conclusion that is both decisive enough to guide action and honest about its margins. The common failure modes are well documented: the primacy effect of the first hypothesis, the anchoring effect of a prior diagnosis in the records, the tendency to average discrepant data rather than investigate the discrepancy, and the tendency to write reports that describe the assessment rather than answer the question.

A workable integration procedure

A defensible procedure has five steps. First, restate the decision the assessment was commissioned to inform. Second, list the hypotheses that were under consideration, including the ones that were discarded. Third, array the evidence relevant to each, noting its source, quality and direction. Fourth, state which hypothesis is best supported and why, and what evidence would have changed the conclusion. Fifth, translate the conclusion into recommendations that are specific, feasible and linked to

identified needs.

This procedure has the additional advantage of being auditable. A colleague reading the record can see which evidence was weighed, which was omitted and whether the conclusion follows. Where the procedure reveals that the evidence is insufficient for any conclusion, that is itself a finding and should be reported as such, with a specification of what further assessment would resolve.

Recommendations that can be used

Recommendations are the part of the report most likely to be acted upon and most likely to be written carelessly. Effective recommendations name an action, an agent, a setting and a review point: "the school to provide extended time in written examinations for the next two terms, reviewed at the end of the spring term with a re-administration of the timed reading measure" is usable; "support with examinations" is not. Recommendations should be proportionate to the evidence, should distinguish what is required from what is desirable, and should not smuggle in clinical claims that the assessment did not establish.

Closing the loop

Assessment ends with feedback and follow-up, not with the signature on the report. Feedback gives the client an opportunity to correct factual errors, to contest interpretations and to influence the plan; follow-up establishes whether the recommendations were implementable and whether the formulation predicted what happened. Both practices improve the accuracy of subsequent assessments, because they expose the assessor to the consequences of their inferences. The ethical warrant for assessment is that it helps; the empirical test of that warrant is whether it did.

Advanced Case Conference

The referral. A community mental health team asks for assessment of a 34-year-old man, referred by his general practitioner, with the question: "Does he have bipolar disorder?" The general practitioner notes two episodes of elevated mood in five years, each following a period of heavy alcohol use and each lasting four to six days; a long history of unstable employment; and a recent presentation with low mood after a relationship ended. The team's psychiatrist has already prescribed a mood stabiliser.

The evidence. Semi-structured interview establishes no episode of elevated mood lasting more than four days, no marked impairment during the episodes, and no hospitalisation. Collateral report from a sibling describes the same two episodes as "him being on form for a week", with no change in functioning. A structured diagnostic interview supports a hypomanic episode; neither episode meets criteria for mania. Screening identifies harmful alcohol use over eight years and a current moderate depressive episode. Records document two earlier diagnoses of recurrent depression, made without assessment of alcohol use.

The reasoning. Three hypotheses compete: bipolar II disorder; alcohol-related mood instability with a superimposed depressive disorder; and a primary depressive disorder with substance use. The structured interview supports the first, but the temporal association with drinking and the absence of inter-episode elevation weaken it. The base rate of bipolar disorder in a community mental health setting is low enough that a single supporting source should not carry the conclusion. Discriminating evidence would be prospective mood charting over three months, corroborated by an informant, together with a period of reduced drinking.

The decision. The report records a provisional diagnosis of bipolar II disorder with a stated confidence level, documents harmful alcohol use as a co-occurring focus of treatment, and recommends prospective monitoring before any change to the mood stabiliser is considered. It states explicitly what would change the conclusion: a third episode independent of substance use, or prospective charting showing sustained elevation with functional change. The general practitioner is advised that the diagnostic question remains open, and that the current treatment plan does not depend on its resolution.

The teaching point. The assessment did not answer the question that was asked, and saying so was the correct outcome. A conclusion that answered it would have required evidence the assessment did not have.

Chapter Summary

Assessment in counselling psychology is a decision process organised around a question, not a battery organised around an instrument. Its products are distinct: diagnosis classifies, formulation explains and treatment planning prescribes, and conflating them produces reports that neither classify accurately nor explain usefully. Interpretation is probabilistic throughout, and base rates, measurement error and method variance determine how much a finding can bear. Functional impairment and context are not ancillary information but part of what a diagnosis means for a particular person. Evidence-based assessment requires instrument selection and threshold setting to be justified by evidence for the specific purpose at hand, and it treats clinical expertise as a discipline of calibration rather than a licence for impression. Finally, the limits of an assessment are part of its conclusion: a report that states its own uncertainty is more useful, and more honest, than one that does not.

Review Questions

1. Distinguish testing from assessment, and explain why the distinction matters for interpreting a score.
2. Using an example from your own placement or service, state a referral question in the form specified in Section 1.3 and identify the decision it informs.
3. Calculate the positive predictive value of a test with sensitivity .85 and specificity .80 in settings with prevalence of 5 per cent and 40 per cent. What follows for screening policy?
4. Explain why a discrepancy between self-report and informant report should be investigated rather than averaged.
5. What does it mean to say that the limits of an assessment are part of its conclusion? Illustrate with a report sentence.

Discussion Questions

1. A court asks you to determine whether a parent has a personality disorder. How do you negotiate the referral question, and what do you record about the negotiation?
2. The team psychiatrist has already started treatment. What are the professional and ethical considerations in reporting a provisional rather than a confirmed diagnosis?
3. Should a diagnosis ever be withheld from a client at the request of a third party? Argue both positions and state your own with reasons.
4. How would you decide whether a new instrument has earned a place in your service's assessment protocol?
5. In what circumstances, if any, is an assessment without standardised instruments defensible?

Further Reading

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PART I

Foundations and Measurement

The purposes of assessment, the logic of clinical inference, interviewing, and the psychometric basis of score interpretation.

Chapter 1 · Chapter 2 · Chapter 3 · Chapter 4

CHAPTER 2

Clinical Interviewing, Mental Status Examination and Diagnostic Reasoning

Learning Objectives

After working through this chapter, the reader should be able to:

- design an interview whose structure follows from the referral question rather than from habit;
- conduct a mental status examination that is systematic, observable and reportable;
- reconstruct symptom chronology and course so that onset, duration and change can be dated;
- evaluate medical, substance-related and developmental explanations before settling on a psychological one;
- apply structured and semi-structured interviews appropriately and understand their limits;
- document diagnostic reasoning in a form that another clinician can audit.

Key terms: presenting concern · history taking · mental status examination · symptom chronology · collateral information · differential diagnosis · structured interview · diagnostic formulation · confirmatory bias · anchoring

Introduction

The interview is the most used and least examined instrument in mental health practice. It is used to establish rapport, gather history, observe behaviour, assess risk and arrive at a formulation, and it is expected to

accomplish all of this within an hour and without a manual. Yet the interview is also a measurement procedure with known properties: it is sensitive to interviewer style, vulnerable to memory reconstruction on the part of both parties, and dependent for its validity on the questions that were asked and the questions that were not.

This chapter treats interviewing as a technical activity that can be designed, evaluated and improved. It sets out how to build an interview from the referral question, how to conduct and record a mental status examination, how to establish chronology, how to test alternative explanations, and how to document reasoning. The emphasis throughout is on the difference between a conversation that produces a warm feeling and an interview that produces defensible evidence. The two are not opposites; the best interviews produce both, but when they diverge, evidence must win.

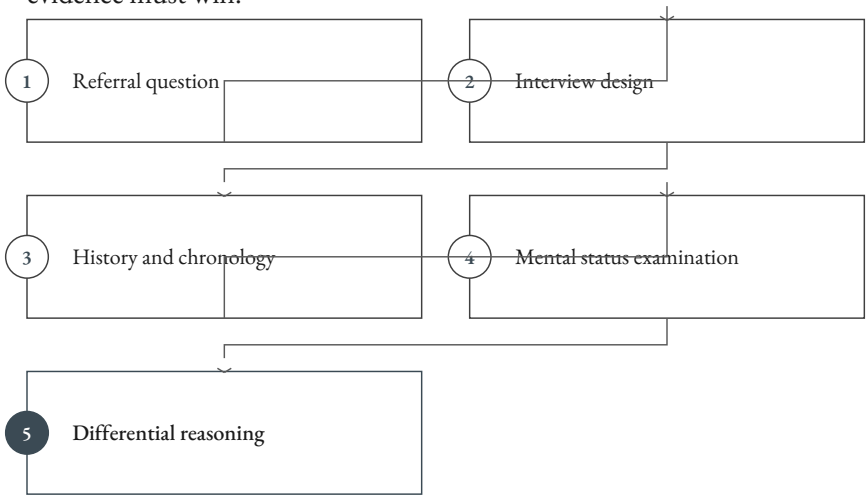


Figure 2.1. The structure of a diagnostic interview. Each element supplies evidence for a different claim, and the mental status examination is interpreted against, not instead of, the history.

2.1 Designing a Purposeful Clinical Interview

An interview should be designed before it is conducted. Design means deciding, in advance, which questions must be answered, which methods

will answer them, how long each element will take and what will be sacrificed if time runs short. Interviews that are not designed tend to follow the client's narrative wherever it leads: informative, often humane, but unreliable as evidence, because the topics that were covered reflect conversational flow rather than the decision at hand.

From question to interview plan

The design process begins with the referral question and works backwards. If the question concerns diagnostic status, the interview must cover criterion-relevant symptoms, their duration and their functional impact, and must probe the boundary with neighbouring conditions. If it concerns formulation for psychotherapy, the interview must establish developmental history, relational patterns, maintaining cycles and the client's own understanding. If it concerns risk, the interview must cover ideation, intent, plan, means, past behaviour, protective factors and the immediate environment. A single interview can serve more than one purpose, but only if the purposes are named and time is allocated to each.

Structure and flexibility

Structure and responsiveness are usually presented as a tension; in practice they are complementary. A structured spine — a fixed sequence of domains that will be covered regardless of how the conversation develops — frees the interviewer to follow the client's account within each domain without fear of losing coverage. The spine protects against the common failure in which an emotionally charged opening topic consumes the session and the developmental history is never taken. Flexibility then operates inside the structure: the order of questions, the depth of probing and the use of silence are all adjustable.

Time budgeting

Experienced interviewers budget time explicitly. A ninety-minute first interview might allocate fifteen minutes to the presenting problem and its chronology, fifteen to developmental and family history, ten to medical and substance history, fifteen to the mental status examination, ten to risk,

ten to the client's goals and expectations, and fifteen to feedback and closure. Where the budget is exceeded in one domain, the interviewer decides consciously what will be deferred and records the decision. Unrecorded omissions become invisible gaps in the eventual report.

The interviewer as an instrument

Every interviewer has a characteristic style and a characteristic set of blind spots. Some ask leading questions under time pressure; some avoid asking about suicide; some over-elaborate the areas they find interesting. Because the interviewer is the instrument, the instrument requires calibration: recording interviews, reviewing them with a supervisor, and comparing one's own conclusions with those of a colleague who saw the same client. These practices are standard in psychotherapy training and under-used in assessment training, where the assumption persists that interviewing is a natural skill rather than a disciplined one.

2.2 Opening the Interview and Establishing Rapport

Rapport in assessment is not a preliminary courtesy but a validity condition. Clients who are anxious, suspicious or uncertain about the purpose of the interview give shorter answers, under-report stigmatised experiences and withhold information about risk. The opening minutes therefore do substantive work: they establish why the assessment is taking place, who will see the results, what the limits of confidentiality are, and what the client can expect to happen.

Explaining purpose and limits

The explanation should be given in plain language and should be specific to the setting. A formulation such as the following covers the essentials without jargon: "You have been referred by your general practitioner because of difficulty sleeping and low mood. I will ask about how this started, what has made it better or worse, and how it is affecting your life. I will write a report for your doctor, and I will show you what I have written

before it is sent. What you tell me is confidential, with the exception of a serious risk to your safety or someone else's, which I would discuss with you first wherever possible."

Consent is then obtained for the assessment as described, and separately for any recording, collateral contact or release of information. Consent obtained at the start is not a substitute for consent at each subsequent stage; clients retain the right to decline particular lines of enquiry.

Managing expectations and anxiety

Clients arrive with assumptions about assessment, often drawn from television, previous negative experiences or fear of being judged. Naming these assumptions early is usually more effective than reassurance. "Some people worry that this is a test they can fail, or that what they say will be used against them. It is not, and it will not be, unless there is a risk I have to act on; if that happens I will tell you." Where a client has had a previous assessment they experienced as dismissive, asking about it yields both rapport and information about how they are likely to present now.

Culture, language and the etiquette of disclosure

The conventions of first meetings vary. Direct eye contact, the use of first names, questions about family or about income may be appropriate in one cultural setting and intrusive in another. Interpreters change the interpersonal geometry of the interview and require their own briefing. Clients from settings where mental distress is expressed somatically may open with bodily complaints, and the interviewer who redirects too quickly to psychological content may lose the account entirely. The practical rule is to ask about, rather than assume, the client's preferences on matters of address, accompaniment and disclosure.

2.3 History of Presenting Concerns

The history of the presenting concern is the evidential centre of the interview. It must establish what the problem is, when it began, how it has changed, what makes it better or worse, what the client has already tried,

and what the client believes is happening. Each of these elements supports a different inference, and each is vulnerable to a characteristic distortion.

Establishing the problem in behavioural terms

Clients describe problems in evaluative language: "I'm a failure", "I can't cope", "my anxiety is terrible". The interviewer's task is to convert these evaluations into descriptions that can be dated, counted and compared. What does "can't cope" look like on a Tuesday morning? How many days in the past month did the client leave the house? What does "terrible" mean on a scale where the worst week of the client's life is ten? Conversion is not pedantry; it is what makes change measurable and what allows a second clinician to understand the baseline.

Onset, duration and course

Dating onset is more difficult than it appears, because clients reconstruct beginnings in the light of present understanding. Three techniques improve accuracy: anchoring to public events (a move, a bereavement, a job change, a pandemic); asking about the last period of normal functioning and working forwards; and obtaining collateral confirmation. The distinction between abrupt and gradual onset, between episodic and continuous course, and between stable and deteriorating trajectory carries substantial diagnostic weight. A first episode of psychosis that develops over two years is a different clinical object from one that develops over two weeks.

Precipitants, maintaining factors and prior attempts

Having established the pattern, the interview turns to what surrounds it. Precipitants are events temporally associated with onset; maintaining factors are processes that keep the problem going, including avoidance, reassurance-seeking, sleep disruption, alcohol use, relationship conflict and occupational strain. Asking what the client has already tried — and what happened — yields information about both the problem and the client's model of it. Clients who have consulted five practitioners and been helped by none are telling the assessor something that should be examined rather

than assumed.

The client's explanatory model

Every client arrives with an explanation. It may be biomedical, moral, interpersonal, spiritual or none of these. Eliciting the explanation serves three purposes: it identifies beliefs that will shape engagement; it may contain information the assessor lacks; and it establishes the terms in which feedback will eventually have to be given. Explanations that are factually incorrect, such as a belief that low mood is caused by a permanently damaged brain, can often be modified through discussion of evidence, but only after they have been heard.

2.4 Developmental, Family and Social History

Developmental history is often deferred when time is short, and its absence is one of the most frequent weaknesses of assessment reports. Yet it supplies information that no current-state measure can: the trajectory of the person's functioning, the presence of longstanding difficulties that were previously tolerated, and the family and social context in which the present problem is embedded.

Developmental milestones and early history

Pregnancy and birth complications, early illness, language and motor milestones, schooling and peer relationships, and the age at which difficulties were first noticed all belong in the record. The purpose is not to assemble a complete biography but to identify deviations with diagnostic relevance: a child whose language developed late and who has always found social rules opaque presents a different differential from one whose difficulties began abruptly at fourteen. Where the client cannot report early history, an informant or contemporaneous record should be sought, and the source of the information stated.

Family history

Family history of mental disorder, substance use, suicide and neurological illness is relevant to both risk and formulation. It should be recorded with the same care as other clinical information: who, what, when, and whether the diagnosis was made clinically or reported by the family. Family history reported by a client who has never met the affected relative is weaker evidence than a documented admission, and the difference should be visible in the record.

Social and occupational history

Education, employment, housing, income, relationships, caring responsibilities and migration history describe the conditions under which the person functions. They are not background colour; they are the context against which impairment is judged. A client who has stopped working because their employer closed has not necessarily deteriorated in functioning, and a client who continues to work while profoundly depressed may be doing so at unsustainable cost. Asking about the shape of an ordinary week is often the most efficient route into this material.

Adverse experiences and trauma

Enquiry about adverse experiences should be routine, specific and paced. Routine, because clients rarely volunteer such material unprompted; specific, because "have you experienced trauma?" invites a considered answer about identity rather than a factual one; paced, because detailed trauma enquiry in a first interview can be destabilising when the purpose is assessment rather than treatment. The interviewer asks what happened, when, and whether it is currently affecting sleep, intrusions or avoidance, and defers detailed narrative to a context where it can be contained.

2.5 Mental Status Examination

The mental status examination (MSE) is the structured description of the client's current psychological functioning, observed and elicited during the interview. Its conventions derive from clinical psychiatry and have been refined over more than a century; its purpose is to make observable data

comparable across examiners and over time. The MSE is not a diagnosis and should not be written as one. It is a record of findings, each of which must be described in terms specific enough to be contested.

Domain structure

A complete examination covers appearance and behaviour, speech, mood and affect, thought form, thought content, perception, cognition, insight and judgement. Table 2.1 lists the domains, the observations that belong to each, and the descriptive vocabulary conventionally used. The order is flexible, but all domains should be addressed, and negative findings should be recorded as explicitly as positive ones: "no perceptual abnormality elicited" is a finding; silence is not.

Table 2.1. The mental status examination: domains and observations

Domain	Observations to record	Example phrasing
Appearance and behaviour	Dress, hygiene, posture, gait, eye contact, cooperation, psychomotor activity	"Dressed appropriately for the season; cooperative; motor activity slowed throughout"
Speech	Rate, volume, prosody, quantity, fluency, articulation	"Normal rate and volume; fluent; no word-finding difficulty"
Mood and affect	Self-reported mood; observed range, reactivity, congruence	"Describes mood as 'flat, grey'; affect restricted but reactive to humour"
Thought form	Rate and flow, coherence, loosening of associations, circumstantiality, perseveration	"Tangential at times; no derailment or incoherence"
Thought content	Preoccupations, obsessions, delusions, overvalued ideas, suicidal and homicidal ideation	"Preoccupied with contamination; no delusional conviction"
Perception	Hallucinations in each modality, illusions, depersonalisation, derealisation	"Denies auditory or visual hallucination; transient derealisation during panic"
Cognition	Orientation, attention, memory, executive function, estimated premorbid level	"Oriented; digit span 6 forwards; recalls 3/3 after delay"
Insight	Attribution of symptoms, acceptance of treatment, understanding of risk	"Attributes difficulties to 'stress' but accepts referral; partial insight"
Judgement	Decision-making, social judgement, risk-related behaviour	"Has continued driving despite blackouts; judgement impaired in this domain"

Observation before enquiry

Much of the examination is observational and should be recorded before the client's self-report is reported. A client who describes severe agitation while sitting still for fifty minutes has supplied a discrepancy worth recording. Affect is observed; mood is reported; the two should never be conflated. The examiner records what was seen, what was heard and what

was said, in language that distinguishes the three: "reported", "appeared", "described".

Cognitive screening within the MSE

A brief cognitive screen belongs in every MSE where cognition is relevant to the question: orientation to time, place and person; attention by digit span or serial sevens; registration and delayed recall of three items; and a measure of executive function such as verbal fluency or a clock-drawing task. Screening is not assessment. A poor screening result indicates the need for formal cognitive assessment, and a good result does not exclude significant difficulty in a high-functioning person, particularly where the screen is affected by education, language or culture. Chapter 6 treats formal cognitive assessment in detail.

Recording and its uses

The MSE should be written in the past tense, in prose, in the third person, and without diagnosis. It should be specific enough that another examiner could compare their own findings on a later occasion, which is the point of a standard vocabulary. Phrases such as "affect appropriate" or "insight poor" should be avoided unless the basis for the judgement is given. The most useful MSE records are those written with an eye to the next assessment rather than only to the present one.

2.6 Symptom Chronology and Course

Chronology is the skeleton of diagnosis. Most of the distinctions that matter clinically depend on sequence and duration: whether low mood preceded or followed heavy drinking; whether obsessions appeared before or after the onset of depression; whether attention difficulties were present before the age of twelve; whether a period of elevated mood lasted four days or four weeks. An interview that establishes the presence of symptoms without establishing their temporal relations has collected half the necessary evidence.

Building a timeline

The most reliable technique is to construct a timeline with the client, working backwards from the present and anchoring each event to something memorable. A sheet of paper with a horizontal line, marked by years and significant events, converts a rambling history into a sequence that both parties can inspect. Discrepancies emerge naturally: the client who dated their low mood to a divorce is found, on the timeline, to have consulted their general practitioner two years earlier.

Duration and episodicity

Duration matters both for diagnostic thresholds and for prognosis. Interviewers systematically under-estimate the duration of episodes and over-estimate the duration of remissions, because recent symptoms are more memorable than recent wellness. Asking "when was the last week in which you felt like your usual self?" is often more informative than asking about the beginning of the current episode.

Course and its diagnostic implications

Course describes the trajectory over time: single episode with recovery, recurrent episodes with inter-episode recovery, continuous course, deteriorating course, or fluctuating course without clear episodes. Course is often more discriminating than symptom content. Rapid onset with full recovery and preserved inter-episode functioning supports a different set of explanations from a slow, insidious deterioration with progressive functional decline.

Reconciling discrepant timelines

Where the client's timeline contradicts the records or an informant, the assessor should not simply prefer one source. Discrepancies usually have explanations: memory decay, shame, different definitions of the problem, or a genuine change in the client's understanding. Recording both accounts and the reason for preferring one is more valuable than silently adopting the more convenient version.

2.7 Collateral Information and Records

Collateral information is evidence supplied by someone other than the client, or documented before the current assessment. Its value is that it is independent of the client's current presentation; its limitation is that it is also independent of the client's current experience. Neither source is intrinsically superior, and the assessor's task is to weigh them.

Obtaining collateral well

Collateral contact requires the client's consent in almost all circumstances, and the client should know what will be asked and what will be shared. Where consent is refused, the assessor can still proceed but should record the refusal and its effect on the confidence of the conclusions. Informants should be briefed on the purpose of the contact, asked for specific observations rather than global judgements, and told what will happen with the information. An informant asked "is he depressed?" will answer with an opinion; an informant asked "what have you noticed about his sleep, appetite and interest in things he used to enjoy over the past two months?" will answer with evidence.

Records: uses and hazards

Records establish chronology, document prior assessments and reveal previous reasoning, including previous errors. They also transmit error with the authority of print: a diagnosis made in a fifteen-minute consultation and repeated in four subsequent letters acquires spurious solidity. The assessor should distinguish primary documentation from summarised reports, note the source and date of each, and treat an unsupported diagnosis in the records as a hypothesis rather than a finding.

Weighing conflicting accounts

Where accounts conflict, the assessor examines three things: the opportunity of each source to observe the behaviour in question; the motive each might have for distortion; and the specificity of the description. A partner who has lived with the client for a decade and

describes nightly waking at three in the morning is a stronger source on sleep than a client who reports sleeping "fine". A teacher who has known the child for six weeks is a weaker source on developmental history than a parent, though possibly stronger on classroom behaviour.

2.8 Differential Diagnosis

Differential diagnosis is the disciplined comparison of competing explanations. It is not the production of a list; it is the identification of the evidence that would separate the items on the list. Performed well, it converts an open-ended interview into a sequence of discriminating questions and gives the eventual report a visible logic.

Generating the differential

The differential should include, at minimum, the most likely explanation, the most serious explanation that must not be missed, and the explanation the client or referrer considers most likely. Neighbouring categories within the same diagnostic family should be considered explicitly, since these are the errors that occur most often: depressive episode versus persistent depressive disorder versus bipolar depression; generalised anxiety versus panic disorder with agoraphobic avoidance; obsessive-compulsive disorder versus generalised anxiety; post-traumatic stress disorder versus adjustment disorder.

Discriminating evidence

For each pair of explanations, the assessor identifies the findings that would separate them: age of onset, episodicity, presence of elevated mood, response to previous treatment, family history, substance involvement, and the presence of a defining feature such as obsessions or flashbacks. These discriminating findings become the priority questions for the remainder of the interview, and their absence from the record is the clearest sign that differential reasoning did not take place.

When the differential cannot be resolved

Some differentials cannot be resolved at a single assessment. Where the discriminating evidence requires observation over time — as it does for bipolar spectrum presentations, for personality difficulties in a young adult, and for the boundary between trauma-related and psychotic phenomena — the correct output is a provisional conclusion with a stated review point, not a forced choice. Recording the unresolved differential, and what would settle it, is the mark of a completed rather than an abandoned assessment.

2.9 Medical and Substance-Related Explanations

A substantial minority of presentations that appear psychological at first contact have a medical or substance-related contributor, and a further group has both. The assessor is not expected to practise medicine, but is expected to know which questions to ask, which findings prompt referral, and how medical and substance factors alter the interpretation of psychological data.

Medical contributors to consider

Conditions that commonly present with psychiatric symptoms include thyroid dysfunction, anaemia, vitamin B12 deficiency, sleep apnoea, chronic pain, diabetes, autoimmune conditions, temporal lobe epilepsy, delirium and early dementia; medications including corticosteroids, beta-blockers, isotretinoin, interferon and some hormonal treatments may also contribute. The interview should establish recent physical investigations, current medication, changes in medication, chronic illness and unexplained physical symptoms. Findings that mandate prompt medical referral include new confusion, focal neurological signs, headache of recent onset with vomiting, unexplained weight loss and first presentation of psychosis in later life.

Substance use

Substance use is systematically under-detected in routine interviews. Standard practice is to ask about tobacco, alcohol, cannabis, stimulants, opioids, prescribed medication used outside prescription, and novel psychoactive substances, using a quantity–frequency approach and a validated screen where time allows. The relevant temporal questions are whether use preceded, coincided with or followed the onset of symptoms, and whether there has been a period of abstinence during which symptoms resolved. A presentation that has never been observed in the absence of substance use cannot be attributed confidently to a primary psychological disorder.

Integrating medical and psychological evidence

Medical and psychological explanations are not competitors in most cases. Chronic illness produces psychological distress; psychological disorder worsens medical outcomes; medication produces symptoms that are then treated as psychiatric. The report should state the medical information available, note what is unknown, and specify what would change the interpretation. Where information is unavailable, the limitation belongs in the conclusions rather than in a footnote.

2.10 Interviewing Under Conditions of Distress

Much assessment occurs when the client is distressed, intoxicated, acutely psychotic, in pain, in crisis or in an emergency department at three in the morning. Under these conditions the standard interview must be adapted, and the assessor must decide what can reliably be assessed now and what must wait.

Acuity and its consequences

Acute distress narrows attention, impairs recall and increases suggestibility. A client in crisis may not be able to give a coherent developmental history, and pressing for one wastes the opportunity to establish safety and engagement. The priority sequence is: immediate safety; physical needs such as pain, hunger, intoxication or withdrawal; orientation and

coherence; the presenting crisis; and only then a fuller history, either later in the same contact or at a scheduled follow-up.

Interviewing clients with psychosis or severe agitation

With psychotic presentations the interviewer adjusts pace, uses shorter sentences, avoids confrontation of delusional content where it is not necessary to the question, and records the content accurately without endorsing it. Agitation calls for environmental measures before verbal ones: reducing noise, ensuring an exit, sitting at an angle, keeping the door open, and involving colleagues where risk is escalating. The interviewer's own arousal is data; noticing fear is a reason to attend to safety rather than to suppress the observation.

Emergency assessment and the limits of consent

In emergencies, assessment may proceed without full consent where necessary to protect life, and the legal basis will vary by jurisdiction. The assessor should record what was done, on what authority, and what the client was told. Capacity to consent to assessment should be considered rather than assumed, particularly where intoxication, delirium, intellectual disability or acute psychosis is present. A short, documented capacity assessment is preferable to a silent assumption either way.

2.11 Structured and Semi-Structured Interviews

Structured and semi-structured diagnostic interviews exist to improve the reliability of diagnosis by fixing the questions, the decision rules and the order of enquiry. Their use is one of the better-evidenced practices in diagnostic assessment, and their neglect in routine practice is one of the more persistent gaps between evidence and service delivery.

The available formats

Fully structured interviews specify every question and every coding decision, allowing administration by trained interviewers without clinical judgement. Semi-structured interviews specify the domains and the

decision rules but leave the wording and the follow-up probing to the clinician. Instruments in common use include the Structured Clinical Interview for DSM-5 (SCID-5; First, Williams, Karg, & Spitzer, 2016), the Mini International Neuropsychiatric Interview (Sheehan et al., 1998) and the ICD-11 based schedules developed for epidemiological and clinical use. Each has a defined training requirement and a defined scope; none assesses everything, and most are weak on personality, risk and functional impairment.

Strengths and costs

The strengths are reliability, completeness and auditability: a structured interview ensures that criterion-relevant questions are asked and that the answers are recorded in a comparable form. The costs are time, training, rigidity and a tendency to produce diagnosis without formulation. The instruments were designed for research and epidemiological use, and their translation into clinical practice requires the addition of exactly the contextual and functional enquiry that they do not supply.

Using them well

Used well, a semi-structured interview forms the spine of the diagnostic part of the assessment, with the assessor supplementing it with history, collateral and functional enquiry. Used badly, it becomes a checklist administered without attention, producing a diagnosis that is technically derived and clinically shallow. The difference is visible in the report: a checklist yields a list of criteria met; a well-conducted interview yields criteria met, alternatives excluded, and the reasoning that connects them.

2.12 Documenting Diagnostic Reasoning

The interview concludes with a written record. Its purpose is to allow another qualified professional to understand what was asked, what was found, what was concluded and how confident the conclusion should be. Documentation is therefore not an administrative afterthought but the final stage of the reasoning process, and it is often the stage at which

weaknesses become visible.

Structure of the interview record

A workable record contains: the source of referral and the question asked; the client's account of the presenting problem; chronological history; developmental, family, social and medical history; substance use; risk; the mental status examination; the methods used, including any structured interview; a formulation; a differential diagnosis with the evidence for and against each alternative; conclusions with confidence; and recommendations. Each section should distinguish reported information from observed information and from the assessor's inference.

Recording uncertainty

Uncertainty should be recorded in specific terms. "Diagnosis uncertain" is uninformative. "Criteria for a moderate depressive episode appear to be met; the assessment of duration depends on the client's dating of onset, which conflicts with the general practitioner's record; a bipolar spectrum explanation cannot be excluded on current evidence" tells the reader exactly where the doubt lies and what would resolve it.

Language, stigma and the client's access

Records are increasingly accessible to clients, and should be written in language that the client could read without harm. Pejorative descriptors, speculation about personality presented as fact, and irrelevant personal detail should be excluded. This is not only an ethical requirement but a clinical one: records written with the client as a potential reader tend to be more precise and less judgemental, and both qualities improve their professional usefulness.

Advanced Case Conference

The referral. A 29-year-old woman is referred by her employer's occupational health service after three months of increasing absence. The stated question is whether she is fit to return to work and whether she has depression.

The interview. She describes low mood, early waking, loss of appetite and loss of confidence beginning nine months earlier after a reorganisation placed her under a manager she experiences as humiliating. She has not worked for eleven weeks. The mental status examination shows slowed speech, restricted but reactive affect, no perceptual abnormality, and intact cognition. She describes passive suicidal ideation without intent. She drinks two bottles of wine most evenings, which she dates to the same period; she has not had a drink-free week since the onset. PHQ-9 score is 18.

The differential. Major depressive episode; alcohol-induced mood disturbance; adjustment disorder with depressed mood; and an underlying anxiety disorder with secondary avoidance. The discriminating questions concern sequence and abstinence: mood symptoms and drinking began within the same month; she reports no period of abstinence long enough to observe mood; her account of "humiliation" includes panic symptoms in the manager's presence dating from before the reorganisation.

The reasoning. The depression is real and severe enough to warrant treatment, and the alcohol use is likely both a consequence and a maintaining factor. The presence of pre-existing panic-like symptoms weakens a pure adjustment explanation and raises the possibility of an anxiety disorder that the reorganisation exacerbated. Fitness for work cannot be determined from severity alone; it depends on the availability of adjustments and on whether the panic can be managed in the specific work setting.

The decision. The report records a moderate depressive episode with harmful alcohol use, notes the unresolved question of an underlying anxiety disorder, and recommends treatment for both before any determination of fitness. It specifies workplace adjustments that would be tested on a phased return, and it states that the assessment of fitness will need to be repeated once drinking has reduced and treatment is established. The occupational health service is advised that returning a diagnosis without a functional assessment would not answer the question it asked.

The teaching point. The interview established sequence, not just severity. The sequence changed the formulation, and the formulation changed the recommendation.

Chapter Summary

The clinical interview is a designed instrument. Its structure should follow from the referral question, its time should be budgeted, and its coverage should be fixed in advance so that emotionally salient material does not displace the domains that matter for the decision. The mental status examination converts observation into a standard vocabulary, and its value depends on the examiner distinguishing what was seen from what was reported. Chronology and course are the backbone of differential reasoning: many diagnostic questions are settled by sequence and duration rather than by symptom counts. Medical and substance-related explanations must be considered before a psychological one is adopted, and structured interviews improve reliability while supplying neither formulation nor functional assessment. Documentation is the final act of reasoning, and a record that states its uncertainty precisely is more useful than one that conceals it in confident prose.

Review Questions

1. Explain how the referral question determines the structure of an interview, using two contrasting referral questions.
2. What is the difference between mood and affect, and why should they be recorded separately?
3. Describe three techniques for improving the accuracy of reported onset.
4. Why does a period of abstinence matter when assessing a presentation complicated by substance use?
5. What does a semi-structured diagnostic interview add, and what does it not add, to a clinical assessment?

Discussion Questions

1. A client refuses consent for collateral contact. How does this affect the confidence of your conclusions, and what do you record?

2. Should every client be asked about suicidal ideation in the first interview? Consider the evidence, the ethics and the practicalities.
3. The records contain a diagnosis made six years ago by a clinician who saw the client once. How do you handle it?
4. How would you adapt an interview for a client who is agitated and suspicious in an emergency department?
5. Is there a place for assessment without a structured interview in doctoral-level practice? Defend your position.

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PART I

Foundations and Measurement

The purposes of assessment, the logic of clinical inference, interviewing, and the psychometric basis of score interpretation.

Chapter 1 · Chapter 2 · Chapter 3 · Chapter 4

CHAPTER 3

Psychometrics: Reliability, Validity, Norms and Measurement Error

Learning Objectives

After working through this chapter, the reader should be able to:

- explain the classical test model and the meaning of observed score, true score and error;
- select and interpret indices of reliability appropriate to the intended use of a measure;
- calculate and apply the standard error of measurement and the standard error of a difference;
- evaluate the five sources of validity evidence set out in current professional standards;
- interpret norms, standard scores and percentiles without overstating their precision;
- reason about sensitivity, specificity and predictive values in relation to base rates;
- identify what a technical manual does and does not establish.

Key terms: observed score · true score · measurement error · reliability coefficient · standard error of measurement · standard error of the difference · validity argument · internal structure · measurement invariance · standard score · percentile rank · sensitivity · specificity · predictive value · cut score · differential item functioning

Introduction

Psychometrics is the discipline that allows an assessor to say how much confidence a score deserves. It is not an optional technical background to clinical practice but the grammar in which score-based claims are made. An assessor who cannot compute a confidence interval around an obtained score, or explain why a difference between two subtests may be unremarkable, is not in a position to interpret the instrument they have used, however well they administered it.

This chapter sets out the minimum technical apparatus required for advanced practice. It treats reliability as a property of scores rather than of tests, validity as an argument rather than a coefficient, and norms as a comparison sample rather than a standard of health. It also insists on a distinction that the clinical literature often blurs: the difference between the accuracy of a measure and the usefulness of a decision made with it. A test can be accurate and still be a poor basis for a decision, if the decision threshold is badly placed or the base rate in the setting is low.

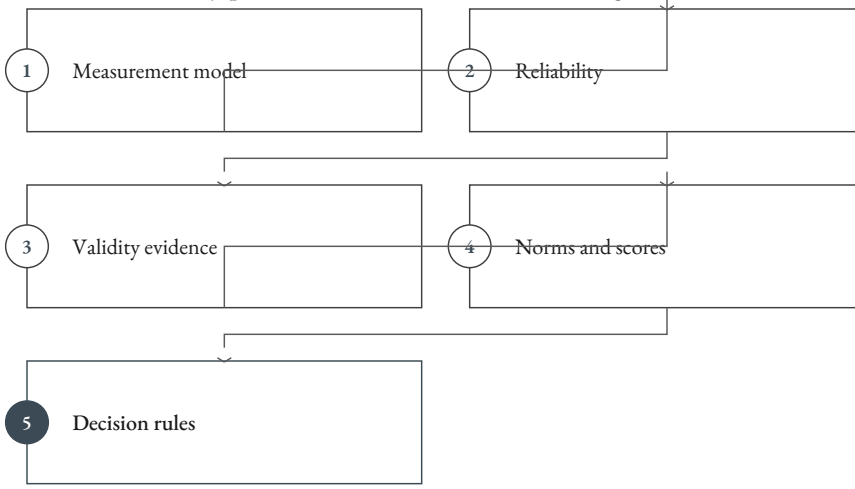


Figure 3.1. The psychometric chain. Each link is a separate empirical claim, and no link can be inferred from the strength of the others.

3.1 Measurement Models and Observed Scores

Every score produced by a psychological test is an observation, not a quantity known without error. Classical test theory formalises this intuition: an observed score is the sum of a true score and an error component, where the true score is defined as the expected value of observed scores across hypothetical repeated administrations. The model is deliberately modest. It does not claim that the true score is a fixed property of the person, only that a given measurement procedure has an expected value for that person at that time.

The classical model and its assumptions

The classical model assumes that error is random, uncorrelated with true score and uncorrelated across measurements. These assumptions are convenient rather than universally true: practice effects, fatigue, mood change and item exposure all produce error that is systematic rather than random. The practical consequence is that classical indices may over-state the stability of a score across occasions, which is why test–retest coefficients are usually lower than internal consistency coefficients for the same instrument.

Domain-sampling and latent-variable models

Two other families of measurement model are worth distinguishing. Domain-sampling models treat a test as a sample from a universe of equivalent items; generalisability theory extends this by allowing the assessor to quantify how much of the error is attributable to particular facets, such as occasion, rater or item set. Latent-variable models treat the observed items as indicators of an unobserved construct and estimate the relationship between them. Item response theory, a latent-variable approach, has the practical advantage of placing persons and items on the same scale, which is why it underpins many modern adaptive tests and norming programmes.

Why the model matters clinically

The model matters because interpretations differ according to which one is assumed. An internal-consistency interpretation treats a high coefficient as

evidence that the items measure one thing; a test–retest interpretation treats it as evidence that the score is stable. Neither implies the other, and neither implies validity. The assessor who asks "what does this coefficient tell me about the claim I am about to make?" will avoid most of the misinterpretations that appear in reports.

3.2 Reliability and Score Consistency

Reliability refers to the consistency of scores across replications of a measurement procedure. It is not a single property but a family of properties, each defined by the source of error that is allowed to vary. A coefficient is meaningless unless the assessor knows which source of error it addresses.

Table 3.1. Forms of reliability and the error each addresses

Coefficient	Source of error allowed to vary	Typical use and caution
Internal consistency (α, ω)	**Item content within a single administration**	**Estimating homogeneity; α assumes unidimensional items and equal loadings**
Split-half with correction	**Item halves within one administration**	**Rarely reported; superseded by α and ω**
Test–retest	**Occasions**	**Stability of scores; affected by practice, memory and genuine change**
Alternate forms	**Item sets**	**Form equivalence; requires large, matched item pools**
Inter-rater	**Raters**	**Consistency of scoring judgements; requires independent ratings**
Inter-method	**Methods**	**Convergence across procedures; lower than within-method coefficients**
Standard error of measurement	All random error	Precision of an individual score; depends on the reliability estimate used

Internal consistency

Cronbach's alpha remains the most reported coefficient, and the most misused. It is a lower bound to reliability under restrictive assumptions of essential tau-equivalence, and it is inflated by test length and by item redundancy. Sijtsma (2009) argued that alpha's usefulness is limited and that its routine reporting has obscured the more relevant question of whether a set of items measures a single attribute; McNeish (2018) and Dunn, Baguley and Brunsden (2014) have made the same case and recommended alternatives such as omega. For the assessor, the practical implications are that a high alpha does not establish unidimensionality, that alpha should be reported for the specific sample used, and that a very high alpha (above .95) in a short scale may indicate redundant items rather than excellent measurement.

Test–retest and stability

Test–retest reliability is relevant whenever a score will be interpreted as a stable characteristic or used to evaluate change. Coefficients are affected by the interval: too short and respondents remember their answers; too long and genuine change attenuates the coefficient. For state measures, low test–retest reliability is expected and appropriate, since the construct itself is meant to vary. Reporting a state measure's stability as though it were a virtue, or a trait measure's instability as though it were a defect, are both category errors.

Inter-rater reliability and agreement

Where a score depends on clinical judgement — a diagnostic interview, a personality rating, a risk assessment — inter-rater reliability is central. Cohen's kappa corrects for chance agreement on categorical judgements and is conventionally interpreted with the benchmarks proposed by Landis and Koch (1977), although these are arbitrary and should be reported alongside raw agreement and prevalence. For continuous ratings, intraclass correlation coefficients are preferable to simple correlations because they account for systematic differences between raters as well as rank-order agreement.

3.3 Standard Error and Confidence Intervals

A reliability coefficient is not directly useful at the level of the individual case. What the assessor needs is an estimate of how far an individual's obtained score is likely to be from their true score. The standard error of measurement provides this: it is the standard deviation of the hypothetical distribution of obtained scores around a person's true score, and it is computed as the standard deviation of the test multiplied by the square root of one minus the reliability coefficient.

Using the standard error

Because the error distribution is approximately normal, a 95 per cent confidence interval is obtained by adding and subtracting 1.96 standard errors from the obtained score. For an instrument with a standard deviation of 15 and a reliability of .90, the standard error is 4.74 and the 95 per cent interval around an obtained score of 100 runs from approximately 91 to 109. Two consequences follow. First, a score is a region, not a point, and reports should present it as such. Second, differences between scores that lie within overlapping intervals should not be interpreted as meaningful.

The standard error of the difference

Comparisons between two scores require a different statistic, because each score carries its own error. The standard error of the difference incorporates the reliability and the correlation between the two measures, and it is always larger than the standard error of either score alone. This is why the difference between a verbal and a performance score of ten points is often unremarkable, and why the interpretation of subtest scatter requires the difference to exceed a threshold that the manual should specify. Assessors who interpret any difference as a profile are reading noise.

Regression to the mean and extreme scores

Where a client is selected for assessment because their score was extreme — a high score on a screening questionnaire, a low score on an intake measure — a second administration will tend to be closer to the mean even if nothing has changed. This is a statistical property, not evidence of improvement or deterioration. Its practical implication is that change should be evaluated against a comparison group or a reliable change index rather than against the raw difference between two scores.

3.4 Validity as Evidence for Interpretation

Validity is the degree to which evidence and theory support the interpretations and uses of test scores. It is a property of an interpretation, not of an instrument, and it is never established once and for all. The current *Standards for Educational and Psychological Testing* frame validation as the accumulation of evidence across five sources: evidence based on test content, on response processes, on internal structure, on relations with other variables, and on the consequences of testing (American Educational Research Association, American Psychological Association, & National Council on Measurement in Education, 2014).

The unified view

Messick's (1995) unified conception of validity integrated these sources into a single argument about score meaning, and it remains the dominant framework. In this view, construct validity is not one type among several but the whole of the argument, of which content, structure and external relations are components. Kane (2013) extended the approach by describing validation as the construction of an interpretive argument that can be evaluated for its inferences and assumptions. Both frameworks share a practical implication: the assessor should be able to state the argument that connects an observed score to the claim being made, and should be able to identify the weakest link in that argument.

What validity is not

Validity is not the same as reliability, though reliability constrains it. It is not established by factor analysis alone, by correlation with another instrument alone, or by the test's widespread use. Nor is it a property that transfers automatically across populations, languages or purposes. An instrument validated for screening in primary care is not thereby validated for forensic determination, and an instrument validated in one cultural setting requires evidence before it is used in another.

Threats to validity in practice

The most common threats in clinical practice are not statistical but procedural: administration that departs from the standard without being recorded; interpretation that ignores the reference group; use of a measure outside its validated age range; coaching or rehearsal of items; and the accumulation of multiple small deviations, none of which alone would invalidate the score. The assessor's protection is documentation specific enough to allow a reader to judge whether the interpretation was warranted.

3.5 Content and Response Process Evidence

Content evidence concerns the relationship between the test's items and the construct domain it is intended to represent. Response process evidence concerns what respondents actually do when they answer. Both are usually established during test development, and both are frequently assumed rather than examined in clinical practice.

Content representativeness

A test's content should sample the domain it claims to measure, in proportions that reflect the domain's structure. This is a judgement informed by expert review, task analysis and, where relevant, curriculum or diagnostic specifications. The relevant question for the assessor is not whether the items look sensible but whether the domain they sample corresponds to the construct in which the referral question is framed. A measure of "depression" that omits anhedonia, or a measure of "executive

function" that samples only one executive subdomain, will produce scores that are valid for something narrower than the label suggests.

Response processes

Response process evidence asks whether respondents are doing what the test assumes they are doing. A reading comprehension item assumes decoding and comprehension; if the respondent cannot decode, the item measures something else. A self-report item about "worry" assumes that the respondent interprets worry in the intended sense, which varies with language, culture and clinical experience. Cognitive interviewing during test development, and think-aloud studies, are the standard methods for examining these assumptions. In clinical practice the equivalent is vigilance: the assessor watches how the client engages with the task, not only what score results.

Fairness as a content issue

Content and response process evidence are also where fairness begins. Items that presuppose knowledge, vocabulary or experience unevenly distributed across groups introduce construct-irrelevant variance, which is a validity problem before it is an equity problem. The professional standards treat fairness as integral to validity rather than as a separate consideration, and this framing is the correct one.

3.6 Relations with Other Variables

Evidence based on relations with other variables examines whether scores behave as the construct implies they should. It includes convergent evidence (correlation with measures of the same or related constructs), discriminant evidence (low correlation with measures of different constructs), criterion-related evidence (prediction of a current or future outcome) and nomological evidence (fit within a network of theoretical relationships).

Convergent and discriminant evidence

A new measure of anxiety should correlate substantially with established anxiety measures and less substantially with measures of, say, working memory. The pattern, not any single coefficient, carries the argument. Where correlations with everything are moderate, the measure may be assessing general distress rather than the specific construct claimed — a common problem in clinical self-report, where negative affectivity inflates correlations among all symptom scales.

Criterion-related evidence

Criterion-related evidence addresses the practical question directly: does the score predict the outcome the decision concerns? For diagnostic instruments, this is expressed as sensitivity, specificity and predictive values, treated in Section 3.9. For prognostic instruments, it may be expressed as the proportion of variance in outcome explained, or as the area under a receiver operating characteristic curve. Criterion evidence is highly context-specific: a coefficient obtained in a specialist clinic may not hold in primary care, because the range of presentations and the base rate differ.

Nomological networks and incremental validity

The strongest validity arguments place a measure within a network of expected relationships derived from theory, including relationships with variables it should *not* predict. Incremental validity is a special and practically important case: does the measure add predictive accuracy beyond information already available, such as a brief interview or an existing record? Many instruments with respectable validity have negligible incremental validity, and the assessor who asks this question is doing more for their clients than the one who asks only whether the instrument is "validated".

3.7 Internal Structure

Internal structure evidence concerns the relationships among items and subscales: whether the measure behaves as a unidimensional scale, a set of

correlated factors or a hierarchical structure with a general factor. It is usually examined by exploratory factor analysis during development and confirmatory factor analysis thereafter.

Dimensionality and its clinical meaning

Dimensionality is not an abstract statistical property. A measure that is unidimensional yields a total score that can be interpreted as a single quantity; a measure with several weakly correlated factors yields a total score that averages different things and may obscure clinically important variation. Many symptom inventories are best modelled as hierarchical, with a general distress factor and specific factors for depression, anxiety and somatic complaints. Interpreting the total score of such a measure as though it indexed a single disorder is a common error with practical consequences for treatment selection.

Factor analysis: uses and limits

Confirmatory factor analysis tests a specified model against observed covariance, and its fit indices are sensitive to sample size, model complexity and the distribution of items. A model that fits adequately in one sample may fail in another, and adequate fit does not establish that the model is the correct one — only that it is not incompatible with the data. Exploratory analysis, conversely, can generate a structure from noise if sample size, extraction method and rotation are chosen carelessly. The assessor reading a manual should look for replication across independent samples rather than a single well-fitting model.

Reliability at the factor level

Where a measure has subscales, reliability should be reported for each. A total score with excellent reliability can contain a subscale with reliability so low that its interpretation is unjustified. Reporting subscale means without subscale reliability is one of the more common defects of clinical reports, and it is particularly consequential when the subscale is the basis for a recommendation.

3.8 Norms, Percentiles and Standard Scores

A norm is a description of the distribution of scores in a defined reference sample. It converts a raw score into a statement about relative standing, and nothing more. The sample defines the comparison, and the comparison is only as meaningful as the sample's relevance to the person being assessed.

Score scales

Standard scores express a person's position relative to the reference sample in units of the sample's standard deviation. Table 3.2 summarises the common scales and their interpretation. Percentile ranks express the proportion of the reference sample scoring at or below the obtained score; they are intuitive but not interval-scaled, so differences between percentiles at the extremes correspond to larger differences in the underlying trait than the same differences near the middle.

Table 3.2. Common score scales and their interpretation

Scale	Mean	Standard deviation	Example and interpretation
z-score	0	1	−1.5: 1.5 SD below the reference mean
T-score	50	10	65: 1.5 SD above the mean, near the 93rd percentile
Deviation IQ / index score	100	15	85: 1 SD below the mean, near the 16th percentile
Scaled score	10	3	7: 1 SD below the mean, near the 16th percentile
Stanine	5	2	3: below average band, roughly the 11th–23rd percentile
Percentile rank	50	—	16: 16 per cent of the reference sample scored at or below this value

The normative sample

Manuals should report the size, recruitment method, date of data collection, stratification variables and exclusion criteria of the normative

sample, together with the reliability of scores in that sample. Three questions follow. Does the sample represent the population to which the client belongs? Is it current enough that secular changes have not invalidated it? And is the sample large enough that the norm tables are stable at the extremes, where clinical decisions often fall?

Age, education and the Flynn effect

Normed scores decline in validity as the reference data age, a phenomenon associated with secular increases in measured ability across successive cohorts (Flynn, 1987). Instruments that are not renormed may therefore over-estimate ability in later cohorts. Age and education corrections improve comparability but can also mask genuine impairment in high-functioning individuals, particularly on cognitive measures where a client's demographic prediction may exceed their obtained score despite real decline. The assessor should know which norms were applied, and should record the basis for any demographic adjustment.

3.9 Sensitivity, Specificity and Predictive Values

Where a score is used to classify people — as above or below a threshold, as likely or unlikely to have a condition — the relevant psychometric properties are those of the classification, not those of the score's correlation with a criterion. The four indices are sensitivity (the proportion of people with the condition who are correctly identified), specificity (the proportion without the condition who are correctly identified), positive predictive value (the proportion of positive results that are true positives) and negative predictive value (the proportion of negative results that are true negatives).

Why predictive values matter more than sensitivity

Sensitivity and specificity are properties of the test; predictive values are properties of the test *in a setting*. As Figure 3.2 shows, for an instrument with sensitivity and specificity of .90, the positive predictive value falls steeply as prevalence falls. At a prevalence of 2 per cent, fewer than one in six positive results is a true positive; at 30 per cent, more than three in four

are. The same instrument, the same cut score and the same patient can therefore yield very different conclusions depending on where the assessment takes place.

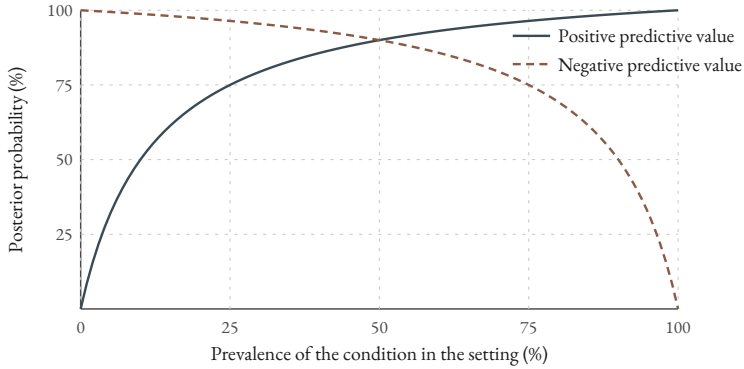


Figure 3.2. Positive and negative predictive values for an instrument with sensitivity .90 and specificity .90, plotted against prevalence. The values are computed from Bayes' theorem; the instrument is held constant while the setting varies.

Likelihood ratios

Likelihood ratios summarise the evidential weight of a result in a form that can be combined with a pre-test probability. The positive likelihood ratio is sensitivity divided by one minus specificity; the negative likelihood ratio is one minus sensitivity divided by specificity. A positive likelihood ratio of 5 raises the odds of the condition fivefold; a negative likelihood ratio of 0.2 reduces them fivefold. Likelihood ratios have the practical advantage of being independent of prevalence, which makes them transportable across settings, and the practical disadvantage of requiring the assessor to estimate a pre-test probability from other information.

From detection to decision

The step from detection to decision requires attention to the consequences of each kind of error. Where a false negative is dangerous and a false positive is manageable, a lower threshold is appropriate; where a false positive triggers irreversible action, a higher threshold is required, together with converging evidence. Naming the asymmetry of consequences is part

of the assessor's duty, because it is the part of threshold setting that no statistic can supply.

3.10 Cut Scores and Classification Accuracy

Cut scores are the operational expression of a decision rule. Their selection is a judgement about the relative costs of false positives and false negatives, informed by the distribution of scores in the relevant populations and by the purpose of the classification.

How cut scores are derived

Cut scores may be derived by criterion-keying (choosing the score that best separates groups defined by an external criterion), by receiver operating characteristic analysis (choosing the point that maximises the balance of sensitivity and specificity, or that satisfies a pre-specified constraint), or by normative convention (choosing a score at a fixed percentile). Each method embeds an assumption about what matters. Criterion-keying depends on the quality of the criterion; ROC analysis depends on the choice of the balance point; normative convention depends on the relevance of the norm group.

Banding and its uses

Because scores carry error, many instruments report interpretive bands rather than exact values. Banding reduces over-interpretation of small differences but can create false discontinuities at band boundaries, where a one-point difference places a client in a different descriptive category. Reporting the obtained score, its confidence interval and the band it falls within is more informative than reporting the band alone.

Classification accuracy in context

Classification accuracy statistics are conditional on the sample in which they were derived. A cut score validated in a sample with a mean score well above the clinical threshold will perform differently in a general population sample. Where local validation is impossible, the assessor should report the

published accuracy statistics, the base rate in their own setting, and the resulting predictive values, and should state that the latter are estimates.

3.11 Fairness and Measurement Invariance

Fairness concerns the extent to which score interpretations are comparable across groups defined by characteristics irrelevant to the construct, such as sex, age, language, culture, disability or socioeconomic status. In psychometric terms, the central question is measurement invariance: whether the instrument measures the same construct in the same metric across groups.

Levels of invariance

Configural invariance requires that the same items load on the same factors in each group. Metric (weak) invariance requires that factor loadings be equal, which permits comparison of correlations and regression coefficients. Scalar (strong) invariance requires that item intercepts also be equal, which is necessary before comparing means. A measure may show configural but not scalar invariance in a given language, in which case group differences in total scores are ambiguous — they may reflect the construct, the response style, or the translation.

Differential item functioning

Differential item functioning occurs when two people of equal standing on the construct have different probabilities of endorsing a particular item because of group membership. It is detected by methods ranging from the Mantel–Haenszel procedure to item response theory based approaches, and its practical remedy is either item revision or interpretation of the scale with the affected items removed. The assessor consulting a manual should look for evidence of invariance across the populations in which the measure will be used, not merely across the whole normative sample.

Translation and adaptation

Translating an instrument is not a matter of linguistic equivalence alone. The International Test Commission's guidelines require documented forward and backward translation, pilot testing, cognitive interviews and evidence of equivalence, together with attention to cultural appropriateness of item content. Where an instrument has been translated without such evidence, its scores should be interpreted with caution, and the limitation should be stated in the report. Chapter 14 extends this discussion to the broader question of culturally responsible assessment.

3.12 Evaluating Technical Manuals

The technical manual is the assessor's primary source for everything in this chapter. Reading it critically is a professional skill, and the criteria are reasonably straightforward once the questions are known.

A reading protocol

A defensible protocol asks the following. What is the construct, and how is it defined? What is the normative sample, when was it collected and how was it stratified? What reliability evidence is reported, for which samples and which score levels? What validity evidence is reported, and against which criteria? Is there evidence of measurement invariance across the groups to which the client belongs? What is the standard error, and how should it be used in reporting? What are the stated limits of interpretation, and what uses does the manual explicitly disclaim?

Common deficiencies

Common deficiencies include norming samples that are large but unrepresentative; reliabilities reported only for the total score; validity evidence confined to correlations with instruments measuring the same construct; absence of invariance evidence for the populations in which the test is marketed; and cut scores presented without their derivation or their accuracy statistics. Reading for these gaps takes less time than reading the marketing material and is considerably more useful.

What the manual cannot tell you

Finally, the manual cannot tell the assessor whether the instrument is appropriate for the particular question, client and setting. That judgement requires clinical knowledge of the presentation, awareness of the service context and attention to the consequences of the decision. The manual constrains and informs that judgement; it does not replace it.

Advanced Case Conference

The referral. A rehabilitation service asks for assessment of a 47-year-old man four months after a moderate traumatic brain injury. The question is whether his reported memory difficulties reflect injury-related impairment or a depressive presentation, and what support he needs on returning to work.

The measures. Two memory measures are administered. The first has a mean of 100 and a standard deviation of 15, with an internal consistency of .93 and a standard error of measurement of 4.0; his score is 84. The second has a mean of 50 and a standard deviation of 10, with a test–retest reliability of .81 and a standard error of 5.6; his T-score is 38. A depression inventory yields a score in the moderately severe range, and a performance validity measure is within normal limits.

The reasoning. The two memory scores are consistent: both lie approximately one standard deviation below the mean, and their confidence intervals overlap substantially. Neither is low enough to suggest severe amnesia. The depression score is high, and the performance validity measure reduces, though does not eliminate, the likelihood that effort accounts for the memory findings. The critical question is whether the memory scores are lower than would be expected from his premorbid level, which requires demographic estimation rather than a comparison with the population mean; his education and occupational history suggest a premorbid level in the above-average range.

The decision. The report states that current memory performance is below estimated premorbid level and below the normative mean, but that the confidence intervals do not permit a precise estimate of the deficit. It records depression as a probable contributor and recommends treatment before a final determination of cognitive impairment. It recommends a phased return with written instructions and reduced memory load, reviewable in three months with repeat testing, and states explicitly that a second assessment will be more informative than the first for estimating change.

The teaching point. The psychometrics did not produce the conclusion; they constrained it. The overlapping confidence intervals prevented an over-interpretation of the difference between two memory scores, and the demographic comparison supplied the question that the norm-referenced comparison could not answer.

Chapter Summary

Reliability and validity are properties of scores and interpretations, not of instruments. Reliability coefficients are meaningful only in relation to the source of error they address, and the standard error of measurement translates a coefficient into a statement about a single client's score. Validity is an argument assembled from content, response process, internal structure, external relations and consequences, and the assessor should be able to identify its weakest link. Norms describe relative standing in a defined sample, and their relevance depends on the sample's correspondence to the client. Sensitivity and specificity describe tests, while predictive values describe tests in settings; the base rate governs the translation, and no instrument can escape it. Finally, fairness requires evidence of measurement invariance, without which group differences in scores cannot be interpreted.

Review Questions

1. Why is a reliability coefficient uninterpretable without knowing which source of error it addresses?
2. Compute the standard error of measurement for a test with a standard deviation of 15 and a reliability of .84, and construct a 95 per cent confidence interval around a score of 92.
3. Explain why the standard error of the difference is larger than the standard error of measurement.
4. Distinguish the five sources of validity evidence in the current professional standards.
5. Using Bayes' theorem, explain why a positive result on a highly specific test may still be more likely to be false than true.

Discussion Questions

1. A colleague reports that an instrument is "validated" because its manual reports a Cronbach's alpha of .94. What questions would you ask?

2. How should an assessor respond when the only available norms for an instrument were collected twenty years ago?
3. What evidence would persuade you that a translated instrument measures the same construct as the original?
4. Should subscale scores be reported when the manual gives no subscale reliability data? Argue both sides.
5. How would you explain a confidence interval to a client in a feedback session without oversimplifying it?

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PART I

Foundations and Measurement

The purposes of assessment, the logic of clinical inference, interviewing, and the psychometric basis of score interpretation.

Chapter 1 · Chapter 2 · Chapter 3 · Chapter 4

CHAPTER 4

Test Selection, Administration, Scoring and Interpretation

Learning Objectives

After working through this chapter, the reader should be able to:

- select instruments by reference to the referral question, the construct and the population;
- evaluate test quality, suitability and the qualifications required for competent use;
- prepare an assessment environment and session that protect the validity of results;
- administer standardised instruments and document any departure from standard procedure;
- apply scoring checks that catch the errors most likely to distort conclusions;
- recognise response styles and validity indicators without over-interpreting them;
- interpret profiles and score patterns conservatively and integrate them with other evidence.

Key terms: incremental validity · test–user qualification · standardised administration · accommodation · scoring reliability · response style · social desirability · validity scale · profile analysis · base-rate error · computer-based assessment

Introduction

Between the decision to assess and the interpretation of a score lies a sequence of practical choices that determine whether the score means anything at all. Which instrument, administered by whom, under what

conditions, scored how, checked against what? These questions are unglamorous and decisive. Most of the serious errors in score-based assessment originate here rather than in the sophistication of the interpretation: an instrument chosen because it was available, administered in an unsuitable room, scored with an outdated norm, and interpreted without attention to the client's response style.

This chapter treats these choices as a connected whole. Selection follows from the question and the population. Administration is a validity condition that can be protected or destroyed. Scoring is a quality-control process with known failure modes. Interpretation begins with what the score is designed to measure, proceeds through a consideration of response style and comparison data, and ends with an integration that draws on evidence outside the test. The chapter closes with the two areas where practice has changed most rapidly: technology-assisted assessment and the recurring errors that experienced assessors make as often as novices.

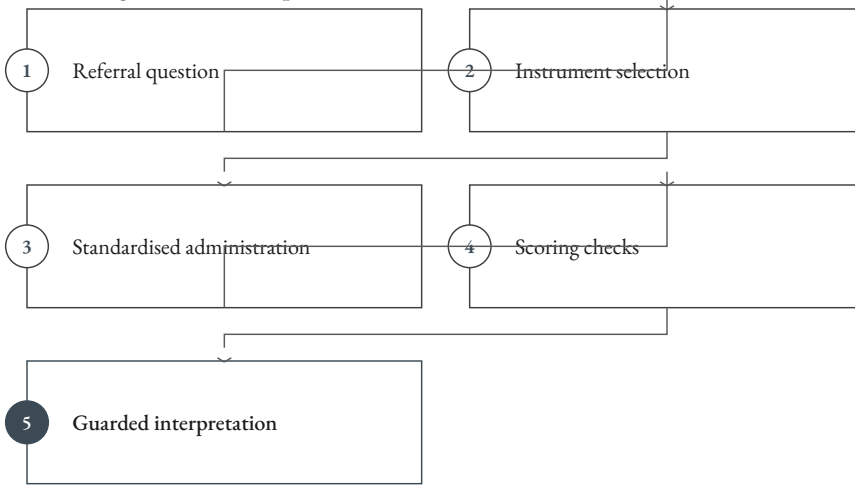


Figure 4.1. The test-use cycle. Selection, administration and scoring each protect or threaten the validity of the interpretation that follows.

4.1 Matching Measures to Referral Questions

Instrument selection is a matching problem: the properties of the measure must correspond to the requirements of the question, the characteristics of the client and the constraints of the setting. When the match is poor, no amount of interpretive skill will recover the assessment.

Specifying the requirement

The first step is to state, in operational terms, what the instrument must deliver. A question about diagnostic status requires a measure with criterion-referenced accuracy data in a comparable population. A question about severity and change requires a measure with demonstrated sensitivity to change and a reliable change index. A question about functional capacity requires a measure of everyday performance rather than symptom report. A question about personality structure requires a measure with evidence on the relevant dimensional or categorical model. Writing the requirement down before consulting catalogues prevents the common pattern in which the available instrument defines the question.

Breadth, depth and sequence

Where several constructs are relevant, the assessor decides the order of enquiry. A common and defensible sequence is broad screening followed by focused measurement: a general symptom inventory to identify elevations, then specific instruments to characterise the domains that matter for the decision. The sequence saves time and reduces the number of instruments administered to every client, but it must be planned so that the screening stage does not determine which hypotheses are entertained. Screening that is used as a hypothesis-generating device is efficient; screening used as a conclusion is not.

Incremental validity and redundancy

Before adding an instrument, the assessor should ask what it will add to what is already known. Two instruments that measure the same construct with similar method variance usually add little; an instrument that measures a different construct, or the same construct by a different method, often adds a great deal. Redundancy is not always a defect —

corroboration has value where the decision is consequential and the constructs are difficult to measure — but it should be a deliberate choice recorded in the rationale.

4.2 Evaluating Test Quality and Suitability

Having identified candidate instruments, the assessor evaluates them against a consistent set of criteria. The criteria are technical, practical and ethical at once, and Table 4.1 sets them out with the question each is designed to answer.

Table 4.1. Criteria for evaluating a candidate instrument

Criterion	Question to answer	Evidence required
Construct definition	Is the construct clearly defined and does it match the referral question?	Manual, theoretical literature
Normative adequacy	Does the reference sample represent this client, and is it current?	Norming chapter; publication date; stratification
Reliability	Are scores sufficiently precise for individual decisions?	Internal consistency, test–retest, inter-rater data
Validity	Is there evidence for the intended interpretation and use?	Content, structure, external relations, consequences
Fairness	Is there evidence of invariance for this client's group?	Invariance and differential item functioning studies
Sensitivity to change	Can the measure detect the change the decision concerns?	Longitudinal studies; reliable change index
Practicality	Is administration feasible in this setting and with this client?	Time, format, literacy, sensory and motor demands
Qualification	Does the assessor hold the training the publisher requires?	Test-user classification; supervisory arrangements
Cost and access	Are materials and scoring affordable and available?	Budget; licensing terms

Evidence beyond the manual

Independent evidence matters, because manuals are published by test developers and are not required to report unfavourable findings. Systematic reviews, independent validation studies and post-publication critiques may reveal limitations that the manual passes over. Where independent evidence is sparse, the assessor should record that fact and calibrate confidence accordingly, rather than treating the manual as a neutral summary of the literature.

Suitability for the individual client

Suitability is a judgement about this person, not about the population. Reading level, visual and hearing acuity, motor control, stamina, language, cultural familiarity with test content and current distress all affect whether the instrument can be administered as designed. Where suitability is marginal, the assessor has three options: choose a different instrument, adapt the administration and document the adaptation as a threat to standardisation, or proceed and interpret with explicit caution. Choosing the third option silently is the error to avoid.

4.3 Qualifications and Scope of Practice

Test publishers classify instruments by the training required for competent use, and professional bodies treat use outside one's competence as an ethical violation regardless of what the publisher permits. The classifications are a floor, not a ceiling: possessing the credential to purchase an instrument does not establish competence to interpret it in a particular context.

Reading the qualification framework

Most frameworks distinguish instruments that require no special qualification, those requiring a professional qualification in psychology or a related field, and those restricted to practitioners with specific training in the instrument and its interpretation. Within the third category, the relevant questions are whether the practitioner has completed the publisher's training, whether they have supervised experience with the

population in question, and whether they maintain competence through continuing professional development and, where relevant, recalibration of scoring judgements.

Boundaries around specialist measures

Certain domains carry boundary requirements that a general doctoral qualification does not satisfy: neuropsychological assessment in medico-legal contexts, forensic risk assessment, assessment for court on parenting capacity, and assessment of capacity to make specific decisions under mental health or capacity legislation. In each case the assessor must know the statutory framework, the reporting conventions and the standards of the relevant specialty. Where the referral requires expertise the assessor does not hold, the correct response is to say so and to contribute what is within competence.

Supervision and delegation

Doctoral trainees administer instruments under supervision, and the supervisor carries responsibility for the interpretation even where the trainee conducted the session. Good supervisory practice includes review of the raw data, independent scoring of a sample of protocols, and discussion of the reasoning that connects scores to conclusions. Where psychometricians or technicians administer tests, the interpreting psychologist remains responsible for selecting the instrument, specifying the conditions of administration, and interpreting the results.

4.4 Preparing the Assessment Environment

The assessment environment is part of the measurement procedure. Noise, interruption, temperature, seating, lighting and the presence of observers all affect performance, and their effects are largest for tasks requiring sustained attention and for clients who are anxious or fatigued.

Physical and interpersonal conditions

A quiet room with a table of appropriate height, seating that allows the assessor to observe without hovering, and freedom from interruption are baseline requirements. The assessor should arrange the room so that test materials are not visible before administration and so that the client cannot see scoring sheets. Where observation by a third party is required — a trainee, an interpreter, a family member — the effect on performance should be considered and, where relevant, measured, since the presence of an observer reliably changes performance on some tasks.

Scheduling and fatigue

Cognitive assessment should be scheduled when the client is likely to be at their best, which for many people is the morning, and should avoid periods immediately after night shifts, during acute pain or in the post-ictal period. Sessions longer than ninety minutes without a break produce fatigue effects that are indistinguishable from impairment. Where a battery requires more time than a single session allows, splitting it across sessions is preferable to compressing it, and the split should be documented.

Preparing the client

Preparation includes explaining what will happen, how long it will take, what the client should do if they need a break, and what will be done with the results. Clients should be told that some tasks will be difficult and that difficulty is expected. They should also be told, in specific terms, what will happen if they decline a task: in most instruments partial administration invalidates the score, and the client has a right to know this before deciding.

4.5 Standardised Administration

Standardisation is the feature that makes norms applicable. The normative sample was tested under specified conditions, and departures from those conditions mean that the comparison is no longer exact. Standardised administration is therefore not bureaucratic rigidity but the condition under which a score can be compared with a norm.

What standardisation requires

It requires fidelity to the manual's instructions on wording, order, timing, materials, prompting, recording and discontinuation rules. It requires that the assessor knows the instructions well enough not to read them mechanically while attending to something else. And it requires that any departure be recorded at the time, with its reason, rather than reconstructed from memory afterwards.

Common departures and their effects

The common departures are familiar: rephrasing a question to be helpful, giving extra time on a timed task, allowing a second attempt, prompting an answer, or continuing past a discontinue rule because the client seems to be improving. Each may be clinically kind and psychometrically costly. A client who is helped to a correct answer has produced a score that reflects the help as well as the ability, and the norm no longer applies. Where an adaptation is clinically necessary, the appropriate response is to record it, to state its likely direction of effect, and to interpret the score qualitatively rather than normatively.

Recording during administration

Contemporary practice records responses verbatim where the instrument requires it, notes latencies and prompts, and records behavioural observations that bear on validity: effort, frustration, guessing, off-task behaviour, and signs of sensory or motor difficulty. These observations often explain discrepancies that would otherwise appear as profile anomalies. Digital recording, where consented to, improves accuracy and supports supervision.

4.6 Accommodations and Accessibility

An accommodation is a change to the standard procedure intended to remove a barrier unrelated to the construct being measured. The distinction between accommodation and modification matters: an accommodation preserves the construct, whereas a modification changes it.

Giving a large-print version of a reading passage to a client with low vision is an accommodation; reading the passage aloud when the construct is reading comprehension is a modification.

Principles

Four principles govern accommodations. They should address a documented functional limitation rather than a diagnosis. They should preserve the construct, or the change should be declared and interpreted. They should be applied consistently across the assessment so that comparisons between scores remain meaningful. And they should be documented in enough detail that a reader could reproduce the conditions.

Common accommodations

Common accommodations include extended time, separate room, rest breaks, large print, screen magnification, sign language interpretation for instructions, written rather than verbal instructions, and alternative response formats such as pointing or dictation. Each has an evidence base of variable quality for the specific instrument, and the assessor should know whether the accommodation has been shown to preserve score meaning for that instrument, or is being applied on general principle.

Documentation and interpretation

Where an accommodation has been applied, the report should state what was changed, why, and how the score should be interpreted as a result. Where no validated norms exist for the accommodated administration, the score should be reported as descriptive rather than norm-referenced, and the assessor should say so. This is more honest than either refusing the accommodation or presenting an accommodated score as though it were standard.

4.7 Scoring and Quality Control

Scoring errors are common, avoidable and consequential. Studies of hand-scored protocols have repeatedly found non-trivial error rates, and

the errors are not random: they cluster where rules are complex, where reversals are required, and where the scorer is tired.

A scoring protocol

A defensible protocol has four elements. Score from the record, not from memory, and record raw responses before applying rules. Apply the manual's rules exactly, including item-level reversal, basal and ceiling rules, and prorating rules where permitted. Score blind to the hypothesis where possible, or have a second person score a sample blind to the expected result. Recompute totals independently, either by hand or with a second entry into the scoring software, and compare.

Checking the output

Automated scoring reduces arithmetic error but introduces its own failures: wrong normative option selected, wrong age entered, wrong test form, wrong scoring key, or software version mismatch. The assessor should verify that the demographic data entered match the client, that the correct norms were applied, and that the output is internally consistent — that subscale scores sum to totals, that percentile and standard scores correspond, and that the reported score falls within the possible range. A short checklist applied to every protocol prevents most errors.

Reporting scores

Reports should give the instrument, the form, the date of administration, the norm group used, the obtained score with its confidence interval and percentile, and a plain-language description of what the score indicates. Where a score is described qualitatively, the descriptor should be tied to a stated range rather than to a general impression: "a score in the low-average range, between the 9th and 24th percentiles" is verifiable; "borderline functioning" is not.

4.8 Response Styles and Validity Indicators

Not every protocol is interpretable. Respondents may under-report symptoms to appear competent, over-report to obtain help or compensation, answer randomly, agree with everything, or misunderstand the task. Instruments with validity scales, and performance validity tests in cognitive assessment, exist to detect these patterns, and their interpretation requires as much care as the interpretation of clinical scales.

The main response styles

Social desirability describes the tendency to present oneself favourably; it is not deception, and it is often adaptive. Malingering describes the intentional production of false or exaggerated symptoms for an external incentive; it is a clinical and legal judgement, not a test result. Over-reporting may also reflect genuine distress, help-seeking, or a personality style characterised by dramatic presentation. Random and acquiescent responding are detectable by inconsistency and infrequency indices.

Interpreting validity indicators

Validity indicators are probabilistic. A single elevated validity scale raises a hypothesis; a pattern of elevations across independent indicators, combined with inconsistent performance on cognitive tasks, raises it further. No indicator establishes malingering, and a normal profile does not exclude exaggeration. The assessor should avoid the two symmetrical errors: dismissing every elevation as "just distress", and treating an elevation as proof of deception.

What follows from an invalid protocol

Where validity is seriously compromised, the appropriate response is to report the finding, describe its probable effect on the scores, and avoid drawing conclusions from the affected data. Where a performance validity test is failed, the interpretation of cognitive scores should be suspended rather than adjusted. In forensic contexts, the finding itself may be material, and the assessor should be familiar with the relevant standards of evidence.

4.9 Interpreting Profiles and Score Patterns

Profile interpretation — the comparison of scores within a battery — is where psychometric discipline most often gives way to narrative invention. Differences between scores are interpreted as evidence of specific strengths and weaknesses, when many such differences are within the range that measurement error and normal variability would produce.

The base-rate problem in profile analysis

The central problem is that patterns are common in the normal population. A two-point elevation on one subscale, a difference of ten points between two indices, or a combination of three moderate scores may occur in a substantial minority of people without any clinical condition. Before interpreting a pattern, the assessor should know its base rate in the relevant population. Where that is unknown, the pattern should be described as an observation rather than interpreted as a finding.

Reliable differences and clinically meaningful differences

A difference must first be reliable (larger than the standard error of the difference) before it can be clinically meaningful. Reliability is necessary but not sufficient: a reliable difference may have no functional consequence. The assessor should ask whether the difference corresponds to something observable in the client's life — a task they avoid, a setting in which they struggle, a report from an informant. Differences that cannot be connected to functioning are usually noise with a narrative attached.

Integrating across instruments

Scores from different instruments should be compared only when the constructs correspond. A high score on a depression inventory and a moderate elevation on a personality scale measuring negative affectivity may reflect the same underlying variance rather than two separate findings. Integration requires a model of what the instruments share, and the assessor should be explicit about which findings are independent and which are likely to be redundant.

4.10 Combining Tests with Other Evidence

Test scores acquire meaning in combination with interview, observation, collateral report and records. The combination is not an average but an argument: each source contributes different information, and the assessor's task is to explain how they fit, where they conflict and which source carries more weight for the specific conclusion.

Convergence, divergence and priority

Where sources converge, confidence rises, provided the sources are genuinely independent. Where they diverge, the assessor investigates rather than averages, and the investigation usually turns on the conditions under which each source was produced: a self-report completed at home during a crisis, an informant report from a relative who sees the client weekly, a test administered after a night without sleep. Specifying those conditions often resolves the discrepancy without any need to prefer one source.

Weighting evidence

Weighting depends on the question. For a question about current symptom severity, a standardised self-report administered recently may carry more weight than an informant's impression formed over months. For a question about lifelong functioning, informant report and records may carry more weight than a self-report completed in an acute state. For a question about capacity to perform a specific task, direct observation of the task is usually the best evidence available.

Presenting integration in a report

Integrated reporting does not place test results in one section and interview findings in another. It organises the report around the questions the assessment was asked to answer, and within each section presents the evidence from all sources that bears on the question. This structure is harder to write and far more useful to the reader, and it makes contradictions visible rather than hiding them in separate sections.

4.11 Technology-Assisted Assessment

Digital administration, online questionnaires, computer-adaptive testing, ecological momentary assessment and automated scoring are now routine. Each offers advantages and introduces failure modes that the assessor must anticipate.

Advantages and risks

Digital administration improves standardisation of timing and presentation, reduces scoring error, enables remote access for clients who cannot attend, and supports repeated measurement. The risks include uncontrolled environments, device variability, connectivity failure, item exposure, and the temptation to administer instruments outside their validated mode. A measure validated in paper format and administered online is not automatically equivalent, and equivalence should be established rather than assumed.

Remote assessment

Remote assessment requires attention to the environment the client is in, the reliability of the connection, the client's familiarity with the technology, and the security of the platform. Cognitive tests that depend on precise timing are particularly vulnerable. Where remote administration is necessary, the assessor should record the conditions and interpret accordingly, and should avoid comparing remote scores with in-person norms unless equivalence has been demonstrated.

Automated interpretation

Automated interpretive reports produce narrative text from score patterns. They are useful as checklists and dangerous as conclusions, because they cannot know the client, the setting or the question, and because their base-rate assumptions are opaque. An assessor who copies an interpretive paragraph into a report without verifying that each claim is supported by the data, and applicable to this client, has transferred responsibility to software. The professional obligation to justify conclusions cannot be

delegated in this way.

4.12 Avoiding Common Interpretation Errors

The recurring errors in test interpretation are well documented and largely preventable. They are not signs of inexperience; several become more likely with confidence.

The principal errors

Confirmatory interpretation: selecting the scores that support the initial hypothesis and discounting those that do not. Base-rate neglect: interpreting a pattern without knowing how common it is. Over-interpretation of small differences: treating scores within the standard error of the difference as meaningful. Reification: treating a label such as "executive dysfunction" as an explanation rather than a description. Context neglect: interpreting a score without reference to the conditions of administration, the client's state, or the demands of their environment. And premature closure: ceasing to gather evidence once a coherent story has formed.

Corrective practices

The correctives are procedural. State the hypothesis before examining the scores and state what would disconfirm it. Compute and report confidence intervals. Consult base-rate tables where available. Write the limitations section before the conclusions section, so that the conclusions must accommodate them. Submit high-stakes conclusions to a colleague for independent review. Keep a record of assessments in which the eventual outcome contradicted the initial interpretation; this is the most effective calibration available to a practitioner.

A discipline of humility

Finally, interpretation should be conducted with explicit awareness of what the instrument cannot show. A test can establish that a person's performance is below that of a reference group; it cannot, by itself,

establish why, whether the difference matters in their life, or what should be done. Those conclusions require the clinical, contextual and ethical reasoning that the rest of this book describes.

Advanced Case Conference

The referral. A university counselling service asks for assessment of a 21-year-old undergraduate who has failed two consecutive years and reports difficulty concentrating. The question is whether a specific learning difficulty is present and whether examination accommodations are warranted.

The assessment. A cognitive battery yields a full-scale score in the average range, with a relative weakness in working memory and processing speed. An achievement battery shows reading accuracy and comprehension in the average range and mathematics in the low-average range. A self-report attention measure is elevated. Performance validity testing is within normal limits. The client reports sleeping five hours a night, working thirty hours a week in addition to study, and describes severe anxiety before examinations.

The reasoning. The profile is common in the general population: a difference between working memory and verbal comprehension of this size occurs frequently and does not by itself indicate a disorder. The achievement scores are consistent with the client's educational history and do not show the discrepancy between ability and attainment that a specific learning difficulty would require. The elevated attention self-report is confounded by sleep restriction and anxiety, and self-report attention measures are known to correlate strongly with distress. The assessment was also conducted during the week before an examination period.

The decision. The report declines to diagnose a specific learning difficulty, explains why the evidence does not support it, and identifies sleep restriction, workload and examination anxiety as the more probable contributors. It recommends a sleep and workload intervention, treatment for anxiety, and a re-assessment after a term if difficulties persist. On accommodations, it recommends limited, function-based provisions — a separate room and rest breaks — while stating that extended time is not supported by the evidence obtained, and that the recommendation may be revisited after reassessment.

The teaching point. The most useful part of the report was the explanation of what the assessment could not establish. A report that had diagnosed a learning difficulty on the basis of an elevated self-report and a common profile would have been plausible, contestable and wrong.

Chapter Summary

Selection, administration, scoring and interpretation form a chain in which each link constrains the meaning of the next. Instruments should be chosen for the question and the population, evaluated against technical and practical criteria, and used only within the assessor's competence. Standardisation is the condition under which norms apply; departures must be documented and their effects stated. Scoring requires independent checks, and automated scoring requires verification rather than trust. Response styles must be considered before clinical scales are interpreted, and validity indicators are probabilistic rather than probative. Profile interpretation should be constrained by base rates and by the standard error of the difference, and integration across sources is an argument rather than an average. Technology changes the mechanics of assessment without changing the professional obligation to justify conclusions in terms of evidence about this client.

Review Questions

1. How does the referral question constrain instrument selection? Illustrate with two contrasting questions.
2. Distinguish an accommodation from a modification and explain why the distinction matters for score interpretation.
3. Describe a scoring quality-control protocol that would catch the three most common scoring errors.
4. Why is a pattern of subtest elevations insufficient on its own to support an interpretation?
5. What are the professional risks of using an automated interpretive report without verification?

Discussion Questions

1. A publisher's classification permits you to purchase an instrument, but you have no supervised experience with the population. What do you do?

2. How should an assessor respond when a client declines a task that is required for a valid total score?
3. Is it ever appropriate to compare scores obtained under different administrative conditions? Under what safeguards?
4. Discuss the ethics of remote cognitive assessment in a rural service with limited alternatives.
5. What would a genuinely independent review of your last ten assessments reveal about your interpretive habits?

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PART II

Assessing Dispositions and Abilities

Personality and interpersonal functioning, cognitive and neuropsychological ability, and achievement, aptitude and career assessment.

Chapter 5 · Chapter 6 · Chapter 7

CHAPTER 5

Personality Assessment and Interpersonal Functioning

Learning Objectives

After working through this chapter, the reader should be able to:

- explain the purposes that personality assessment serves in counselling psychology;
- compare trait, dimensional and categorical models of personality and their clinical implications;
- select broad-band and narrow-band inventories for specific referral questions;
- assess interpersonal and attachment functioning using multiple methods;
- evaluate response styles and impression management in self-report personality data;
- appraise the evidence for performance-based approaches and their appropriate use;
- communicate personality findings without reification or stigma.

Key terms: trait model · five-factor model · person–environment fit · broad-band inventory · narrow-band measure · interpersonal circumplex · attachment style · impression management · validity scale · personality functioning · level of personality functioning · reification

Introduction

Personality assessment asks a different kind of question from symptom assessment. Symptoms describe what is happening now; personality describes the relatively stable organisation of dispositions, motives, strategies and relational patterns that shape how a person responds across

situations and over time. Because that organisation is inferred rather than observed directly, personality assessment depends more heavily on multiple methods and on cautious inference than any other area covered in this book.

The clinical value of the enterprise is considerable. Personality information improves prediction of treatment response and alliance, illuminates the interpersonal cycles that maintain distress, guides the choice between supportive and interpretive approaches, and provides the client with a description of enduring patterns that is often experienced as more respectful than a diagnostic label. The risks are equally considerable: personality language lends itself to reification, to moral judgement disguised as description, and to conclusions that follow the client for years. This chapter develops both the technical and the ethical dimensions of the work.

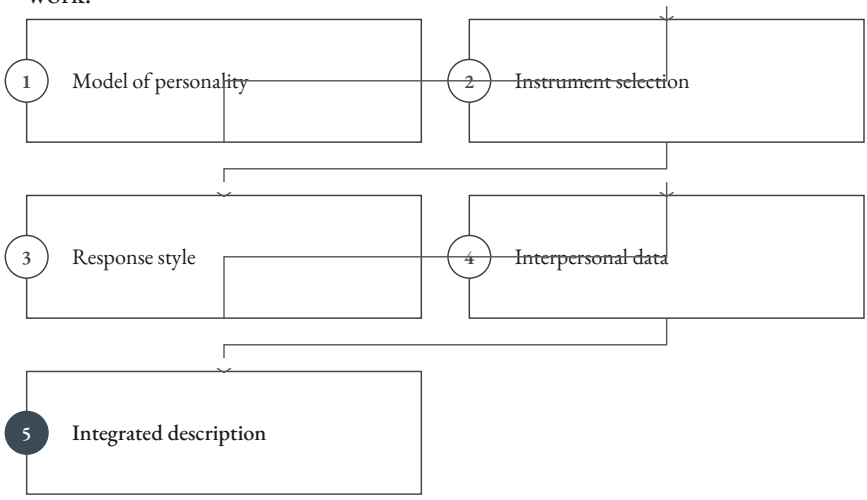


Figure 5.1. Personality assessment integrates model, method and context. No single instrument supplies the construct; the assessor must construct it.

5.1 Purposes of Personality Assessment

Personality assessment is commissioned for several distinct purposes, and the purpose determines the method, the depth and the appropriate form of

reporting. Confusing the purposes produces reports that answer a question nobody asked.

Differential diagnosis and formulation

In diagnostic assessment, personality measures help to distinguish enduring patterns from current states, to identify traits that predispose to particular presentations, and to characterise the personality context in which a symptom disorder occurs. For formulation, the relevant question is usually functional: which enduring strategies maintain the presenting problem, and what would a treatment that addresses them look like?

Treatment planning and prediction

Personality data inform treatment planning in several ways. They predict alliance and dropout risk, identify the interpersonal behaviours likely to be enacted with the therapist, suggest the level of interpretive work the client can tolerate, and indicate the likely response to structured versus exploratory modalities. These predictions are probabilistic and should be stated as such; the evidence for personality-based treatment matching is stronger for some dimensions — such as trait conscientiousness and adherence, or interpersonal hostility and alliance rupture — than for others.

Selection, fitness and legal questions

Personality assessment is also used in employment selection, fitness-for-duty evaluations, custody proceedings and forensic determinations. These applications raise the stakes and change the requirements: the assessor must be familiar with the relevant legal standards, must use instruments with evidence of validity for the specific purpose, must evaluate response style rigorously, and must be able to withstand cross-examination. Instruments validated for clinical description are not thereby validated for forensic prediction, and the distinction should be recorded in the report.

5.2 Trait and Dimensional Models

The dominant framework for describing individual differences is the trait model, in which people are positioned on continuous dimensions along which they differ in degree rather than in kind. The five-factor model, comprising neuroticism, extraversion, openness to experience, agreeableness and conscientiousness, has accumulated substantial cross-cultural and longitudinal support and provides the reference structure for most contemporary inventories.

Strengths of the trait approach

Traits are measurable with high reliability, stable over years in adulthood, heritable to a moderate degree, and predictive of important outcomes including occupational performance, relationship stability and mental health. The five-factor structure has been recovered in many languages and cultures, though the adequacy of the model in every cultural setting remains an empirical question rather than a settled fact. For the assessor, the trait model's principal virtue is that it describes without explaining away: it can say that a client is high in neuroticism and low in agreeableness without implying pathology.

Limits of the trait approach

Traits describe average tendencies, not behaviour in a specific situation. A person high in conscientiousness may be punctilious at work and chaotic at home, and the trait score will not reveal which. Traits also say little about the organisation of personality — how conflict is managed, what happens under stress, how the person experiences themselves and others — which is precisely the information that psychodynamic and interpersonal traditions supply. Finally, extreme trait scores are not diagnoses: high neuroticism is a risk factor, not a disorder.

Hybrid and severity models

Contemporary models attempt to combine trait description with a measure of severity or impairment. The alternative model for personality

disorders in DSM-5-TR specifies both a level of personality functioning, rated on dimensions of self and interpersonal functioning, and a set of maladaptive trait domains; ICD-11 characterises personality disorder primarily by severity with optional trait qualifiers (American Psychiatric Association, 2022; World Health Organization, 2024). Both approaches reflect the evidence that traits alone do not distinguish disorder from variation, and that severity of impairment carries most of the clinical information.

5.3 Broad-Band Personality Inventories

Broad-band inventories assess multiple domains within a single administration. They are the workhorses of clinical personality assessment and the instruments most likely to be encountered in doctoral training.

The Minnesota Multiphasic Personality Inventory

The MMPI family, most recently the MMPI-3 and the MMPI-2-RF, combines clinical scales with extensive validity scales and has an unusually large research base (Ben-Porath & Tellegen, 2008). Its strengths are empirical scale construction, sophisticated validity indices and strong normative data. Its limitations include length, reading level, the interpretive complexity of its higher-order structure, and a normative sample that, while large, is specific to the United States. The instrument is best used as a hypothesis-generating device: its scales identify domains in which further enquiry is warranted, and its validity scales inform the interpretation of everything else.

The Personality Assessment Inventory

The Personality Assessment Inventory (PAI; Morey, 2007) was constructed with a coherent theoretical model, uses a four-point response format, and includes validity scales, clinical scales and treatment consideration scales. Its treatment scales are particularly useful in counselling psychology, since they address treatment rejection, aggression, suicidal ideation and stress. Its reading level is lower than the MMPI's,

which extends its applicability, though not to clients with significant literacy limitations.

Five-factor inventories

The NEO Personality Inventory-3 (McCrae, Costa, & Martin, 2005) and comparable five-factor instruments provide a normal-range description of personality with excellent psychometric properties. They are often the most useful instruments for counselling psychology, because they describe the person rather than the pathology, and because their facet-level scores connect to specific behaviours. Their limitation is that they were not designed to detect disorder, and their clinical utility depends on the assessor's ability to translate trait descriptions into functional implications.

Table 5.1. Representative personality instruments and their principal uses

Instrument	Model and structure	Principal use	Key caution
MMPI-3 / MMPI-2-RF	Empirical, hierarchical; validity and clinical scales	Broad clinical description; response style detection	Length; interpretive complexity; US norms
Personality Assessment Inventory	Theoretical; validity, clinical, treatment scales	Clinical description with treatment focus	Requires clinical integration; limited cross-cultural evidence
NEO-PI-3	Five-factor; facets	Normal-range description; formulation	Not designed to detect disorder
PID-5	DSM-5-TR maladaptive trait domains and facets	Dimensional personality disorder assessment	Trait description without severity
SCID-5-AMPD	Level of functioning plus traits	Hybrid personality disorder diagnosis	Interviewer training and time
Interpersonal circumplex measures	Circumplex; dominance and affiliation	Relational patterns and interpersonal problems	Circumplex assumptions; limited coverage of intrapsychic processes

5.4 Narrow-Band Measures

Narrow-band measures assess a single construct or a small cluster of related constructs in depth. They are useful where the broad-band profile has raised a specific question, and they are often better validated for that specific question than the broad instrument that generated it.

Selecting a narrow-band measure

Selection follows the same criteria as any other instrument: construct definition, normative adequacy, reliability at the score level used, validity for the intended interpretation and evidence of fairness for the client's group. The additional consideration is specificity: a measure of "perfectionism" should have evidence that it distinguishes perfectionism from conscientiousness and from obsessive-compulsive symptoms, or the assessor will be measuring general distress under a more specific name.

Examples and their uses

Common narrow-band measures in counselling psychology include measures of defence style, emotion regulation, anger, shame, self-esteem, dependency and self-criticism, alexithymia, and interpersonal problems. Each has a research literature connecting it to treatment process; the assessor should know that literature before using the instrument to make a treatment recommendation. A measure of defence style, for example, is most useful when the assessor can say what a particular defence implies for the therapeutic relationship, not merely that the client scores high on it.

Combining narrow- and broad-band data

The combination is most informative when the narrow-band finding contradicts or qualifies the broad-band profile. A client with a modest elevation on a broad depression scale but an extreme score on a narrow measure of self-criticism may be presenting a picture that the broad instrument smooths over. Discrepancies of this kind are the most productive starting points for formulation.

5.5 Interpersonal and Attachment Assessment

Personality is expressed in relationships, and the assessment of interpersonal functioning is often more clinically useful than the assessment of traits. Two traditions dominate: the interpersonal circumplex, which describes behaviour along dimensions of dominance and affiliation, and attachment theory, which describes expectations about the availability and responsiveness of others.

The interpersonal circumplex

Circumplex measures position a person's characteristic interpersonal behaviour on a circle defined by dominance (agency) and affiliation (communion). The resulting profile describes, for example, a person who is warm and submissive, or cold and dominant, and it identifies the interpersonal problems that accompany each position — being too controlling, too distant, too self-sacrificing or too vindictive. The model's clinical value lies in its account of complementarity: dominant behaviour invites submission, and hostile behaviour invites hostility, which is how interpersonal patterns become self-maintaining.

Attachment assessment

Attachment is assessed by interview, by self-report questionnaire and by observation of interaction. The Adult Attachment Interview (George, Kaplan, & Main, 1985) classifies attachment states of mind from the coherence of narrative about early relationships; self-report measures such as the Experiences in Close Relationships questionnaire (Brennan, Clark, & Shaver, 1998) assess attachment anxiety and avoidance. The two traditions measure different things — one, the organisation of discourse about attachment; the other, self-reported orientation in close relationships — and they do not always agree. The assessor should know which is being measured and should not treat one as a proxy for the other.

Clinical use

Interpersonal and attachment data are most useful when they are translated into predictions about the therapeutic relationship and into hypotheses about recurrent relational patterns. "The client describes intense worry about abandonment and responds to perceived criticism with withdrawal; in the assessment sessions this appeared as silence following a clarifying question" is a formulation that can be worked with. A circumplex coordinate alone is not.

5.6 Response Style and Impression Management

Self-report personality data are always affected by the respondent's approach to the task. In clinical settings, the two most consequential styles are under-reporting, often driven by social desirability, and over-reporting, which may reflect genuine distress, help-seeking, or deliberate exaggeration.

Social desirability and defensiveness

Social desirability is normally distributed in the population and is not in itself pathological. At moderate levels it improves the accuracy of some predictions, since people who present themselves favourably also tend to behave favourably. At extreme levels it invalidates self-report, because the respondent is answering a question about who they would like to be. The assessor's response is to seek corroboration and to interpret the profile as a description of the client's self-presentation rather than of the client.

Over-reporting and its interpretations

Elevated scores on over-reporting indices may indicate deliberate exaggeration, a cry for help, a personality style characterised by dramatic expression, or simply severe distress in a person with limited vocabulary for it. Distinguishing among these requires collateral information, consistency across measures, and attention to the pattern of elevations. The assessor should resist both the naïve reading (the client is exaggerating) and the defensive reading (the client is distressed), and should record the evidence for whichever interpretation the data support.

Interview-based response style

Response style also operates in interviews. Clients present differently to different interviewers, emphasise different aspects of their history depending on what they believe the interviewer wants, and adjust their accounts in the light of the interviewer's reactions. Structured interviews reduce but do not eliminate this. The interviewer's protection is the same as elsewhere: multiple sources, recorded observations, and explicit attention to discrepancies.

5.7 Performance-Based and Projective Approaches

Performance-based measures present ambiguous stimuli and score the responses according to a standardised system. They have a long history in clinical psychology, a contested evidence base, and a continuing role in assessment, particularly where the assessor needs information about implicit processes or where self-report is compromised.

Evidence

The evidence for performance-based measures is uneven and depends on the specific variable being scored. Mihura, Meyer, Dumitrascu and Bombel (2013) conducted systematic reviews and meta-analyses of Comprehensive System Rorschach variables and found that some variables have good validity while others do not; the valid variables tend to concern perceptual accuracy and thought disturbance rather than the broad personality inferences for which the instrument is best known. This finding supports the use of specific Rorschach indices for specific purposes, and it does not support the interpretive traditions that read personality from response content.

Appropriate use

Performance-based measures are most defensible when used to answer a specific question for which a specific variable has demonstrated validity, as one source among several, by an assessor trained in the scoring system and aware of its limits. They are least defensible when used as a general

personality description, when scored impressionistically, or when their output is granted more authority than the self-report data it contradicts.

Narrative and collaborative uses

Thematic apperceptive methods retain a place in assessment for eliciting the client's narrative themes, conflicts and representations of relationships, particularly in therapeutic assessment models. Their use in this mode is idiographic: the assessor is not scoring against norms but listening for themes that can be discussed with the client. This is a legitimate clinical activity, provided it is not presented as a standardised finding.

5.8 Personality Functioning and Severity

The most important development in personality assessment over the past two decades is the separation of personality variation from personality disorder by introducing a dimension of severity. Severity describes how far a person's personality organisation impairs their capacity to function in the domains of self and relationships; it is what distinguishes a person with difficult traits from a person with a personality disorder.

Domains of personality functioning

The DSM-5-TR alternative model specifies two broad domains, self and interpersonal, each with two elements. Self functioning comprises identity (the experience of a coherent sense of self and the capacity to regulate self-esteem) and self-direction (the capacity to pursue meaningful goals). Interpersonal functioning comprises empathy (the capacity to understand and appreciate others' experience) and intimacy (the capacity for enduring, mutually satisfying close relationships). ICD-11 uses a similar structure, grading severity from mild personality difficulty through moderate disorder to severe disorder.

Assessing severity

Severity is assessed by interview, by informant report and by observation of the client's functioning over time. Useful indicators include the stability of

the client's self-concept, the coherence of their narrative, their capacity to maintain relationships and employment, the quality of their reflection on their own contributions to difficulties, and the presence of self-harm or repeated crises. Structured instruments, such as the Level of Personality Functioning Scale, support the judgement but do not replace it.

Why severity matters clinically

Severity predicts treatment intensity, risk, and the likelihood of dropout. A client with mild personality difficulties may benefit from time-limited work in a general service; a client with severe personality disorder requires a longer, more structured approach with explicit attention to alliance and containment. Confusing the two leads to either over-treatment or, more commonly, to under-treatment disguised as a diagnosis.

5.9 Personality Assessment in Case Formulation

Personality findings become clinically useful when they are translated into a formulation: an account of how the person's enduring patterns interact with current circumstances to produce and maintain the presenting problem.

From trait to pattern to process

The translation proceeds in three steps. Traits become patterns when they are described in terms of recurring behaviour across situations. Patterns become processes when they are described in terms of mechanism: what triggers them, what they achieve, and what they cost. A client high in neuroticism and low in agreeableness may show a pattern of anticipating criticism and responding with counter-criticism; the process is a cycle in which anticipatory vigilance produces behaviour that elicits the criticism that confirms the expectation.

Developmental and relational context

Formulation situates the pattern in its developmental history and its current relational field. The relevant questions are how the pattern

developed, what function it served, and what maintains it now. A pattern that developed as a protection in an unpredictable household may persist long after the household has gone, maintained by its own consequences. This account is explanatory, testable and non-judgemental in a way that a trait description is not.

Writing the formulation

A written formulation of personality should describe the pattern, its origins, its maintaining processes, its costs and its adaptive value, in language the client could read without feeling accused. It should distinguish what is known from what is inferred, and it should identify the point at which intervention is most likely to be effective. Where the formulation cannot be written in such language, it is usually not yet a formulation.

5.10 Feedback from Personality Assessment

Feedback is where personality assessment either becomes useful or becomes harmful. Clients are frequently given personality labels without explanation, and the experience of being described in a report they never see, in terms they would not recognise, is a recognised source of harm.

Principles of feedback

Feedback should be collaborative, specific and behavioural. It should avoid trait labels used as explanations, translate findings into examples the client recognises, invite correction, and connect each finding to something the client can act on. Where the assessment identified a pattern the client had not noticed, the feedback should present it as a hypothesis to be tested rather than a verdict.

Discussing difficult findings

Some findings are difficult: low empathy, high hostility, marked dependency, or a pattern of self-defeating behaviour. The evidence suggests that these findings can be discussed productively when they are framed as

protective or adaptive in origin, described in behavioural terms, connected to the client's own stated goals, and accompanied by explicit acknowledgement of the assessor's limits. Feedback that moralises, or that uses the assessment as an opportunity to say what the assessor has always thought, is harmful and should not be delivered.

Follow-up

Feedback should be followed by an opportunity to respond, and the client's response should be recorded. Where the client disputes a finding, the dispute is information: it may indicate that the interpretation was wrong, that it was correct but premature, or that it was correct and unwelcome. Distinguishing among these requires the assessor to remain curious rather than defensive.

5.11 Cultural Interpretation of Personality Scores

Personality inventories were developed largely in North America and Western Europe, and their item content, normative data and implicit models of healthy functioning carry cultural assumptions. The assessor must therefore consider whether the construct, the items and the norms are appropriate for the client.

Cultural variation in trait expression

The distribution of traits differs across cultural groups, and the meaning of a given trait level may differ more than the distribution. Assertiveness, emotional restraint, deference to family and directness of self-disclosure are valued differently, and a score that is average in one setting may represent unusual conformity or unusual deviation in another. The five-factor structure has been broadly replicated, but the interpretation of specific facets is more culturally variable than the factor structure suggests.

Norms, reference groups and comparison

Where norms are available for the client's cultural or linguistic group, they should be used, with attention to their quality. Where they are not, the

assessor should interpret scores with explicit caution and should prefer within-person and behavioural comparisons. A useful practice is to ask the client what their own reference group would consider typical, which often supplies context that no norm table contains.

Avoiding cultural attribution of pathology

There is a persistent risk that culturally patterned behaviour is read as personality pathology: restraint as constriction, deference as dependency, indirectness as evasiveness, or emotional expressiveness as histrionic. The corrective is not to exempt cultural behaviour from assessment but to generate cultural explanations as competing hypotheses and to test them with the client, the family and, where appropriate, a cultural consultant.

5.12 Integrating Personality Findings

Integration requires the assessor to combine self-report, interview, informant data, behavioural observation and, where used, performance-based measures into a description that is coherent, evidence-based and useful. The obstacles are predictable: sources disagree, instruments measure overlapping constructs, and the pressure to produce a definitive description is strong.

A procedure for integration

A workable procedure begins with the questions the assessment was asked to answer, not with the instruments. For each question, the assessor arrays the evidence from each source and notes its direction and quality. Where sources converge, the finding can be stated with confidence. Where they diverge, the assessor examines the conditions under which each was obtained and decides which is more informative for the specific question. The final description should state the pattern, its functional consequences, its probable developmental origins and its implications for treatment.

Writing without reification

The most important stylistic discipline in personality reporting is to avoid turning a score into an entity. "The client is a narcissistic personality" is a reification; "the client describes a strong need for admiration and reacts to criticism with rage, which has cost him two jobs and a marriage" is a description. The second can be discussed, tested and treated; the first can only be accepted or rejected. Reports should describe behaviour, its function and its consequences, and should use diagnostic terms only where a diagnosis has been formally established and is required by the recipient.

Limits of the assessment

Finally, the report should state what the assessment could not establish: the stability of the pattern over time where only one assessment was conducted, the possibility of state effects on trait measurement, the absence of informant data where none was available, and the limits of the norms for this client. A personality assessment that acknowledges these limits is more credible than one that presents a fully formed portrait on limited evidence.

Advanced Case Conference

The referral. A psychotherapy service asks for assessment of a 34-year-old woman referred after two failed courses of treatment, the first ending in abrupt termination and the second in a crisis admission. The question is what kind of treatment, if any, is likely to help.

The evidence. Interview reveals longstanding difficulty in close relationships, a history of intense but unstable attachments, recurrent self-harm in her twenties and a pattern of idealisation followed by devaluation of therapists. A five-factor inventory shows very high neuroticism, low agreeableness and low conscientiousness. A broad clinical inventory shows elevations across several scales with a validity profile suggesting moderate over-reporting. Self-report attachment measures indicate high anxiety and high avoidance. There is no informant data; the client declined collateral contact.

The reasoning. The severity of impairment in self and interpersonal functioning, combined with the longstanding pattern and the repeated treatment ruptures, is consistent with a moderate to severe personality disorder. The trait profile is consistent but non-specific, and the over-reporting suggests that the self-report scores should be interpreted as an indication of distress rather than as precise estimates. The pattern of rapid idealisation and devaluation has direct implications for treatment: an unstructured exploratory therapy is likely to reproduce the rupture, while a structured approach with explicit agreements about contact between sessions, a clear treatment frame and a named clinician responsible for coordination is more likely to hold.

The decision. The report describes the pattern and its severity, states the diagnosis provisionally, and recommends a structured, formulation-driven treatment with explicit frame agreements and a named coordinator. It states what would indicate that the treatment is working, and what would indicate that it should be reviewed. It records the absence of collateral data as a limitation and the client's reasons for declining it.

The teaching point. The assessment's value lay in what it said about the treatment, not in what it said about the personality. The most consequential finding was a pattern of relationship rupture, and it changed the recommendation.

Chapter Summary

Personality assessment describes enduring organisation rather than current state, and it requires multiple methods and cautious inference. Trait models provide a reliable descriptive structure but do not by themselves distinguish disorder from variation; severity of impairment in self and interpersonal functioning carries most of the clinical information. Broad-band inventories supply comprehensive description with validity scales, narrow-band measures supply depth, and interpersonal and attachment frameworks supply the relational context in which personality difficulties are enacted. Response style affects every self-report measure and must be interpreted rather than ignored. Performance-based measures are defensible where specific variables have demonstrated validity for specific questions, and not as general personality description. Integration should proceed from the question, describe patterns and processes rather than traits-as-entities, and state the limits of what a single assessment can establish.

Review Questions

1. Distinguish trait, dimensional and severity-based approaches to personality and state what each contributes to diagnosis.
2. What does a validity scale measure, and what can it not establish?
3. Explain the difference between what the Adult Attachment Interview and a self-report attachment measure assess.
4. Why is severity of impairment more informative than trait extremity for treatment planning?
5. Describe three ways in which a personality report can reify a score into an entity, and how to avoid each.

Discussion Questions

1. A client's self-report suggests marked over-reporting, but the clinical interview suggests genuine and severe distress. How do you proceed, and what do you record?

2. Should performance-based measures be used in doctoral-level counselling psychology assessment? Defend a position with reference to the evidence.
3. How would you give feedback about low empathy without damaging the therapeutic relationship?
4. What are the risks of using personality assessment in employment selection, and what safeguards would you require?
5. The client declines all collateral contact. How does this change the confidence of a personality formulation?

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PART II

Assessing Dispositions and Abilities

Personality and interpersonal functioning, cognitive and neuropsychological ability, and achievement, aptitude and career assessment.

Chapter 5 · Chapter 6 · Chapter 7

CHAPTER 6

Cognitive, Intellectual and Neuropsychological Assessment

Learning Objectives

After working through this chapter, the reader should be able to:

- identify the questions that cognitive assessment can and cannot answer;
- compare major theories of intellectual functioning and their implications for interpretation;
- evaluate cognitive profiles using base rates, discrepancy standards and estimated premorbid ability;
- assess attention, memory, executive function, language and visuospatial skills by domain;
- select and interpret performance validity measures appropriately;
- integrate medical, neurological and developmental context into cognitive conclusions;
- communicate cognitive findings in functional terms that clients and referrers can use.

Key terms: intelligence · g factor · Cattell–Horn–Cattell model · process overlap · premorbid estimation · discrepancy analysis · processing speed · working memory · executive function · performance validity · ecological validity · cognitive reserve

Introduction

Cognitive assessment addresses a distinctive class of questions: what can this person do with their intellectual resources, how does that capacity

compare with what would be expected for them, and what does the difference imply for their daily functioning, education or work? These questions matter in developmental assessment, in neurological and psychiatric conditions affecting cognition, in educational and occupational decisions, and in legal contexts where capacity or injury is at issue.

The technical demands are high. Cognitive measures have the most elaborate normative structures in psychology, the strongest evidence on reliability, and the most developed methods for estimating change. They are also the measures most vulnerable to misuse: a low score can be read as a diagnosis, a discrepancy as a deficit, and a profile as an explanation, when none of these inferences is warranted without context. This chapter sets out the theory, the methods and the interpretive discipline required to make cognitive findings both accurate and useful.

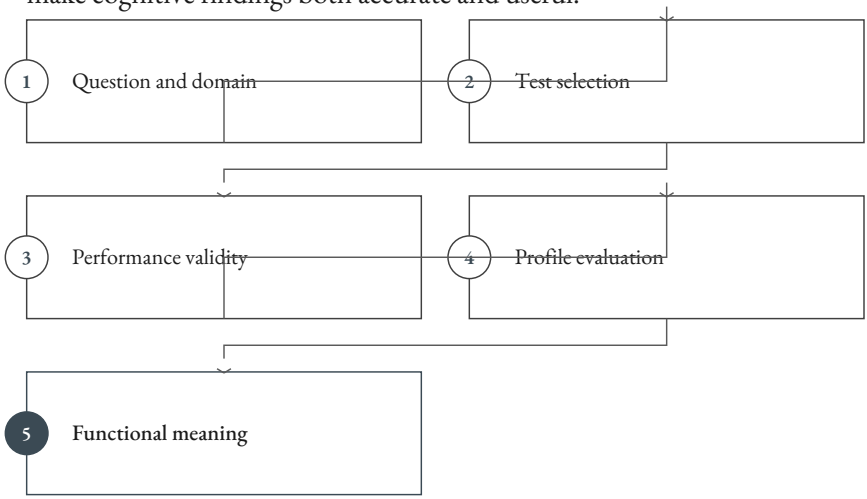


Figure 6.1. Cognitive assessment as a sequence of claims, each requiring its own evidence. Capacity, context and consequence are separate questions.

6.1 Questions Addressed by Cognitive Assessment

Cognitive assessment answers four broad types of question, and the assessor should know which is being asked before selecting an instrument.

Descriptive questions

Descriptive questions concern the person's current level of performance relative to a reference group: how does this child's reading accuracy compare with others of the same age; how does this adult's processing speed compare with the general population? Descriptive questions are the easiest to answer and the most often confused with the other three.

Diagnostic and aetiological questions

Diagnostic questions ask whether a recognised condition is present: intellectual disability, a specific learning disorder, attention-deficit/hyperactivity disorder, a neurocognitive disorder, or an acquired brain injury. Aetiological questions ask what caused an observed impairment. Cognitive tests contribute to both, but rarely settle either alone: a test score is compatible with many causes, and diagnosis usually requires history, course, medical investigation and functional evidence.

Change and monitoring questions

Change questions ask whether functioning has altered: has cognition declined since a previous assessment; is treatment affecting attention; is rehabilitation producing recovery? These questions require instruments with demonstrated sensitivity to change, reliable change indices, and either a previous assessment or a well-justified estimate of premorbid functioning.

Functional and decision questions

Functional questions ask what the person can do in their own environment, and decision questions ask what should follow: what support is needed at school, whether the person can manage their affairs, whether they can return to work. Cognitive tests inform these questions but do not answer them directly. The relationship between test performance and everyday functioning — ecological validity — is moderate at best, and the assessor must supplement test data with behavioural and informant evidence.

6.2 Theories of Intellectual Functioning

Interpretation of cognitive scores presupposes a model of what intelligence is. Three families of model dominate, and each carries different implications for how a profile should be read.

The general factor

Spearman's observation that performance on seemingly diverse cognitive tasks is positively correlated led to the concept of a general factor, *g*, which accounts for a substantial proportion of variance across tasks. The finding has been replicated many times and remains one of the most robust in differential psychology. Its interpretive implication is conservative: because all cognitive tasks share variance with *g*, differences between tasks are smaller than they appear, and a profile of strengths and weaknesses may reflect measurement noise more than cognitive structure.

Hierarchical models

The Cattell–Horn–Cattell model proposes a hierarchy in which *g* sits above broad abilities — fluid reasoning, crystallised knowledge, visual–spatial processing, auditory processing, short-term working memory, long-term storage and retrieval, and processing speed — which in turn sit above narrow abilities. This model underpins most contemporary intelligence tests, and it supplies the vocabulary in which profiles are described. Its practical contribution is the identification of broad domains that have distinct developmental trajectories, distinct neural substrates and distinct implications for learning.

Alternative and process accounts

Alternative accounts emphasise processing rather than structure. The process overlap theory of Kovacs and Conway (2016) explains positive correlations among diverse tasks through the overlap of executive processes they recruit, rather than through a single underlying capacity; other accounts emphasise working memory capacity, relational processing or neural efficiency. These models are not yet the basis of clinical instruments,

but they matter for interpretation: they caution against treating a broad-ability score as a thing, and they direct attention to the processes that produce performance.

Implications for the assessor

The practical implications are threefold. First, broad scores should be reported and interpreted before narrow ones, because they are more reliable and account for more variance. Second, profile differences should be interpreted only where they exceed the standard error of the difference and where there is corroborating functional evidence. Third, the choice of model should be stated, because instruments built on different models produce scores that are not directly comparable.

6.3 Interpreting Cognitive Profiles

Profile interpretation is the most contested practice in cognitive assessment. The idea that a pattern of relative strengths and weaknesses identifies a specific condition has intuitive appeal and limited empirical support.

The discrepancy approach and its limits

Discrepancy analysis compares a person's attainment or domain scores with an estimate of their general ability. The approach is used in the identification of specific learning disorders, where the defining feature is attainment substantially below the level predicted by ability, and in the identification of cognitive decline, where current performance is compared with estimated premorbid functioning. Its limitations are well known: the comparison is between two fallible estimates, so the standard error of the difference applies to the discrepancy as much as to either score; the base rate of discrepancies in the general population is high; and the decision about how large a discrepancy must be is a policy choice rather than an empirical discovery.

Estimated premorbid functioning

Estimating premorbid ability is necessary whenever decline is suspected. Three methods are available: demographic estimation from age, sex, education and occupation; reading-based estimation using tests of irregular word reading, which are relatively resistant to neurological decline; and best-performance estimation using the highest score obtained across the history. Each has limitations: demographic estimates are imprecise at the individual level; reading-based estimates are invalid in people with pre-existing reading difficulties, non-native speakers or limited education; and best-performance estimates assume that the person's best performance was genuinely their optimum.

Base rates and the clinical significance of profiles

The base-rate problem is acute here. A difference of one standard deviation between verbal and visuospatial ability, or between a full-scale score and a single index, occurs in a substantial proportion of the healthy population. Before interpreting such a difference as a finding, the assessor should consult published base-rate tables for the instrument and should ask whether the pattern corresponds to something observable: a task the person avoids, a domain in which they have always struggled, or a functional consequence that others have noticed.

6.4 Attention and Processing Speed

Attention is not a single capacity but a family of processes: alerting and arousal, sustained attention or vigilance, selective attention, divided attention and attentional control. Processing speed describes the rate at which cognitive operations are performed. Together they underpin almost every other cognitive task, which is why their assessment must precede the interpretation of domain-specific scores.

Distinguishing the components

Sustained attention is assessed by continuous performance tasks requiring detection of infrequent targets over time. Selective attention is assessed by

tasks requiring suppression of distractors, such as Stroop-type procedures. Divided attention is assessed by dual-task paradigms. Processing speed is assessed by simple and choice reaction time tasks, symbol substitution and timed coding. Because these components are separable, a general complaint of "poor concentration" requires specification before it can be assessed.

Confounds

Performance on attention tasks is exquisitely sensitive to sleep, pain, medication, anxiety, depression, motivation and substance use. A single assessment conducted during a period of sleep restriction cannot distinguish an attention disorder from the effects of sleep restriction, and the assessor should establish sleep and substance history before interpreting any attention score. Effort and engagement must be evaluated with performance validity measures, because attention tasks are the easiest to underperform deliberately.

Reporting attention findings

Reports should specify which component was impaired, under what conditions, with what confidence, and with what functional consequence. "Sustained attention was reduced relative to estimated premorbid functioning; the client reports difficulty reading for more than ten minutes and has stopped driving on motorways" is a finding with implications. "Attention deficits" is a label without one.

6.5 Learning and Memory

Memory assessment distinguishes encoding, storage and retrieval, and separates verbal from visual material and immediate from delayed recall. Recognition procedures distinguish retrieval failure from storage failure, and interference paradigms examine susceptibility to proactive and retroactive interference.

Domains and methods

Verbal learning is typically assessed by list-learning tasks that measure acquisition across trials, recall after interference, and recognition. Visual memory is assessed by reproduction or recognition of figures. Working memory is assessed by digit span, spatial span and complex span tasks that require simultaneous storage and processing. Prospective memory — remembering to do something in the future — is often more relevant to everyday functioning than retrospective memory and is often omitted from batteries.

Interpreting memory scores

Three interpretive questions structure the analysis. Is the memory difficulty disproportionate to the person's overall cognitive level, or part of a general decline? Is it material-specific, affecting verbal but not visual memory, which may suggest lateralised dysfunction and warrants medical referral? Is it a retrieval problem, in which case cues improve performance, or an encoding problem, in which case they do not? Each question has implications for both aetiology and rehabilitation.

Functional relevance and everyday memory

Laboratory memory tests predict everyday memory complaints weakly. The assessor should therefore supplement them with self-report and informant measures of everyday memory, and with behavioural observation during the assessment, such as whether the client remembers instructions across tasks. Where test performance and everyday functioning diverge, the divergence is itself a finding that requires explanation, often in terms of strategy use, environmental support or emotional factors.

6.6 Executive Functions

Executive functions are the supervisory processes that organise goal-directed behaviour: inhibition, shifting, updating of working memory, planning, monitoring and emotional regulation. They are among the most ecologically important cognitive functions and among the most

difficult to measure.

The measurement problem

The central difficulty is that executive function is defined by the demands of a task rather than by its content, and structured test settings supply exactly the structure that executive function is required to provide in everyday life. A client may perform normally on a formal planning task in a quiet room while being unable to organise a shopping trip. This is the "executive paradox": laboratory measures of executive function correlate modestly with everyday executive behaviour, and informant ratings often predict real-world outcomes better than tests do.

Assessment strategy

A defensible strategy combines three sources: performance-based tests of specific executive components; informant-rated questionnaires of everyday executive behaviour, completed by someone who knows the client well; and observation of the client's behaviour during the assessment itself, including their ability to follow multi-step instructions, to shift between tasks, and to manage frustration. The convergence of these sources, or the absence of convergence, is the finding.

Reporting executive findings

Because executive difficulties are most visible in unstructured, novel or emotionally demanding situations, reports should describe the conditions under which the difficulty appears rather than stating a global deficit. This has direct implications for recommendations, which should focus on external structure, environmental modification and routines rather than on the client's effort.

6.7 Language and Visuospatial Skills

Language and visuospatial ability are the two broad domains most often lateralised and most often affected differentially by neurological events. Their assessment contributes both to diagnosis and to rehabilitation

planning.

Language

Language assessment covers naming, fluency, comprehension, repetition, reading and writing. In counselling psychology settings, the most common reasons for assessing language are suspected neurodevelopmental language disorder, suspected aphasia after stroke or injury, and the need to establish whether a client can comprehend an instrument administered in the test language. Brief measures of verbal fluency and naming are sensitive to a range of conditions and are worth including in most cognitive batteries.

Visuospatial and constructional skills

Visuospatial assessment covers visual perception, spatial judgement, construction, visual memory and, in some batteries, navigation and neglect. Copying tasks, block design, judgement of line orientation and spatial span are standard methods. Difficulties may reflect primary visual impairment, which should be excluded by a vision check, or constructional apraxia, or a broader non-verbal learning difficulty.

The verbal–non-verbal discrepancy

The discrepancy between verbal and visuospatial performance is commonly interpreted as evidence of a non-verbal learning disability or of right-hemisphere dysfunction. The interpretation is weak when based on a single administration and strong only when the discrepancy is large, stable, corroborated by functional evidence and consistent with history. Where the discrepancy reflects an artefact of language dominance in a multilingual client, the error can be substantial.

6.8 Neuropsychological Screening

Not every assessment requires a full neuropsychological battery. Screening instruments exist to identify clients who need fuller assessment, to monitor change, and to answer simple questions efficiently.

Screening versus assessment

A screen is a brief procedure with high sensitivity and moderate specificity, designed to ensure that no significant case is missed. It is not designed to characterise a profile or to establish a diagnosis. Instruments such as the Montreal Cognitive Assessment (Nasreddine et al., 2005) and the Mini-Mental State Examination (Folstein, Folstein, & McHugh, 1975) are widely used screens; both are affected by education, language and culture, and both have limited sensitivity to executive and visuospatial impairment in high-functioning people.

When to refer

Referral for full neuropsychological assessment is indicated where a screen is abnormal, where cognitive complaints are accompanied by functional decline, where a neurological condition is suspected, where a decision with significant consequences (capacity, fitness, guardianship) depends on cognitive status, and where the client's presentation is inconsistent with the available information. The assessor should state the reason for referral explicitly, since a vague referral produces a vague assessment.

6.9 Performance Validity

Performance validity refers to the extent to which a person's performance on cognitive testing reflects their true ability. It is distinct from response bias on self-report questionnaires, and it must be assessed in any context where secondary gain is plausible and, increasingly, in all cognitive assessments as a matter of routine.

The rationale

Performance validity tests are designed to be failed by almost no one with genuine cognitive impairment of the severity being assessed, but to be failed by a substantial proportion of people who are deliberately underperforming. Their logic rests on forced-choice paradigms, in which the probability of scoring below chance can be calculated, and on embedded indices within standard tests that detect improbable patterns of

performance.

Interpretation

Interpretation requires attention to the base rate of invalid performance in the relevant setting, which ranges from near zero in routine clinical samples to substantial proportions in medico-legal and disability evaluation contexts. A single failed measure is generally insufficient; current consensus statements recommend multiple indicators and consideration of the overall pattern (Sweet et al., 2021). Importantly, invalid performance does not establish malingering: it establishes only that the scores cannot be interpreted as estimates of ability.

What the assessor must report

Where validity is compromised, the assessor must state which measures were failed, the base rate of such failure in comparable samples, the probable effect on other scores, and the limits this imposes on any conclusion. Failing to report an invalid performance, or interpreting cognitive scores despite it, is a serious professional error.

6.10 Medical and Neurological Context

Cognitive findings cannot be interpreted apart from medical and neurological information. The same score carries different implications depending on whether it was obtained before or after a neurological event, during or between episodes of illness, on or off medication, and in the presence or absence of pain and sleep disturbance.

Conditions that shape cognitive findings

Relevant conditions include traumatic brain injury, stroke, epilepsy, multiple sclerosis, Parkinson's disease, dementia and other neurodegenerative conditions, HIV, thyroid disease, vitamin deficiencies, chronic pain, sleep apnoea, and the cognitive effects of psychiatric disorders themselves. Medications with cognitive side effects are common, particularly anticholinergics, benzodiazepines, anticonvulsants and some

antidepressants. Substance use, both acute and chronic, affects nearly every domain assessed.

Integrating medical information

The assessor should obtain the medical history, current diagnoses, medication list, results of neuroimaging and other investigations where available, and the chronology relating medical events to cognitive change. Where such information is unavailable, the report must say so, and conclusions about aetiology should be correspondingly cautious. The boundary of competence is important: the identification of a neurological condition is a medical judgement, and the psychologist's contribution is the characterisation of cognitive and functional consequences.

Medical red flags

Certain findings require prompt medical referral rather than psychological formulation: acute confusion, new focal neurological signs, recent-onset seizures, severe headache with vomiting, rapid cognitive decline, and cognitive change in the context of fever or systemic illness. The assessor should know these and act on them without delay.

6.11 Functional Implications of Cognitive Findings

The value of a cognitive assessment to the client depends on its translation into functional implications. This is a separate step of reasoning, and it is the step most often skipped.

From score to activity to participation

A useful translation moves through three levels. The score describes performance on a task. The task maps onto an activity of daily living, such as managing medication or following a conversation in noise. The activity maps onto participation, such as living independently or sustaining employment. Each mapping requires evidence, ideally from observation or informant report, and each introduces uncertainty that should be stated.

Ecological validity and its limits

Ecological validity research shows moderate relationships between cognitive test performance and everyday functioning, stronger for some domains than others. Memory and executive measures tend to predict instrumental activities better than processing speed or simple attention. The relationship is improved by using multiple measures, by including informant ratings, and by assessing the person in or near their own environment.

Recommendations that follow

Functional recommendations should be specific to the activity and the setting, and should name the support rather than the deficit: a written medication schedule and a weekly check with a pharmacist, rather than "needs support with medication"; a reduced-distraction workspace and written instructions, rather than "has attentional difficulties". Recommendations should also indicate what would prompt review, since cognitive function and environmental demands both change.

6.12 Communicating Cognitive Results

Cognitive reports are read by clients, families, teachers, employers, insurers and tribunals. Each audience needs the same core information in a different form, and the assessor's task is to make the report accurate, accessible and honest about its limits.

Reporting scores

A score should be reported with its name, its scale, its percentile and its confidence interval, together with a plain-language descriptor. Reports should make clear what the score measures and what it does not: a working memory index does not measure intelligence, and a full-scale score does not measure potential. Where normative comparisons are uncertain because of language, culture, education or sensory factors, the report should say so at the point where the score is presented, not in a final paragraph.

Explaining uncertainty

Clients and referrers often prefer certainty, and assessors often supply it. The professional obligation is to explain uncertainty in terms the recipient can use. "Your scores on these tasks were lower than we would expect given your education and previous work, but the difference is within the range that could be produced by the fatigue and pain you described. I would like to repeat the assessment when you are sleeping better, because that will tell us more than a single test can." This is honest, respectful and actionable.

Feedback as part of the assessment

Feedback on cognitive findings should include the client's own experience of the tasks, the assessor's observations, the functional implications and the recommended next steps. Where findings are unwelcome, the assessor should allow time for reaction and should avoid presenting conclusions as fixed. And where the client's account differs from the assessment, the difference should be explored, since the client often holds information that the test room could not supply.

Advanced Case Conference

The referral. A 58-year-old man is referred by his employer's occupational health service after increasing errors at work over eighteen months. His job requires managing multiple projects and remembering detailed specifications. The question is whether he has a cognitive impairment and whether he can continue in his current role.

The assessment. He obtains a full-scale score in the average range. Verbal comprehension is high average, perceptual reasoning average, working memory low average and processing speed borderline. Delayed memory is low average, with a large primacy effect and poor free recall that improves substantially with cueing. Executive measures show reduced planning efficiency and increased perseveration. Performance validity is within normal limits. A depression inventory is in the moderate range; he reports sleeping five hours and waking early since his wife's death two years ago. He takes a prescribed antihistamine nightly and drinks two to three glasses of wine most evenings.

The reasoning. The profile is not consistent with a generalised dementia: verbal knowledge is preserved, memory shows a retrieval pattern rather than a storage failure, and validity is adequate. It is consistent with a depression-related attentional and executive difficulty, compounded by sleep restriction, anticholinergic medication and alcohol. The memory pattern is the key discriminator: cortical memory disorders typically show poor encoding and limited benefit from cueing, whereas the pattern observed here is more typical of reduced effortful retrieval.

The decision. The report describes the profile, declines to diagnose a neurocognitive disorder, and identifies depression, sleep and substance factors as probable contributors requiring treatment before any conclusion about dementia. It recommends a review of the antihistamine with his general practitioner, treatment for depression, and reassessment in six months. On fitness for work, it recommends temporary redistribution of the most memory-demanding projects, with a review, and states that a determination of permanent capacity would be premature.

The teaching point. The most important finding was a pattern, not a score. The pattern directed attention to reversible contributors, and the recommendation was to treat them before concluding anything about permanent decline.

Chapter Summary

Cognitive assessment answers descriptive, diagnostic, change-related and functional questions, and the assessor must know which is being asked. Theories of intelligence supply the reference structure for interpretation, and hierarchical models are the most useful clinically. Profile interpretation is constrained by the standard error of the difference, by base rates in the general population and by the need for functional corroboration. Attention, memory and executive function each require domain-specific assessment and attention to confounds; executive function in particular is poorly captured by laboratory tasks alone. Performance validity must be evaluated in every cognitive assessment and reported when compromised. Medical and neurological information is essential to aetiological conclusions, and referral is mandatory when red flags are present. The final step, translation into functional implications, is what makes the assessment useful.

Review Questions

1. Distinguish descriptive, diagnostic, change-related and functional questions in cognitive assessment.
2. Why should broad-ability scores be interpreted before narrow ones?
3. Describe three methods of estimating premorbid ability and one limitation of each.
4. What is the executive paradox, and what does it imply for assessment strategy?
5. What does a failed performance validity test establish, and what does it not establish?

Discussion Questions

1. A parent requests a cognitive assessment to "prove" that their child needs a special school placement. How do you negotiate the referral?
2. How should an assessor respond when a cognitive profile is inconsistent with the client's reported functioning?

3. Is it ever appropriate to interpret a cognitive profile in the absence of premorbid data? Under what conditions?
4. What are the ethical issues in assessing cognitive function for a medico-legal report where secondary gain is plausible?
5. How would you explain a borderline processing speed score to a client who fears dementia?

Further Reading

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PART II

Assessing Dispositions and Abilities

Personality and interpersonal functioning, cognitive and neuropsychological ability, and achievement, aptitude and career assessment.

Chapter 5 · Chapter 6 · Chapter 7

CHAPTER 7

Achievement, Aptitude, Career and Functional Assessment

Learning Objectives

After working through this chapter, the reader should be able to:

- distinguish achievement, aptitude, adaptive behaviour and functional capacity as constructs;
- assess reading, writing and mathematics difficulties in a way that informs instruction;
- evaluate aptitude measures and understand their predictive limits;
- use interest and values inventories in career assessment without determinism;
- assess adaptive behaviour and functional capacity through multiple informants and methods;
- contribute psychological evidence to disability and accommodation decisions;
- write functional recommendations that services and employers can implement.

Key terms: achievement · aptitude · curriculum-based measurement · learning disorder · adaptive behaviour · functional capacity · work personality · interest–environment fit · reasonable accommodation · support needs · transferable skills

Introduction

The assessments described in this chapter share a focus on what a person can do: academically, occupationally and in the activities of everyday life. They are frequently commissioned for decisions with immediate material

consequences — examination access arrangements, special educational provision, disability benefits, occupational fitness, vocational rehabilitation — and they are therefore subject to a particular kind of pressure. The assessor is asked not only what the evidence shows but what conclusion would help.

Counselling psychologists encounter this work in educational settings, in occupational health, in vocational rehabilitation and in disability determination. The technical demands differ from clinical assessment, but the underlying discipline is the same: constructs must be defined, instruments matched to the question, norms checked for relevance, and conclusions tied to evidence. What is distinctive is the centrality of functioning and environment. A score describes a capacity; a decision requires an account of the demands the person faces and the support that would allow them to meet those demands.

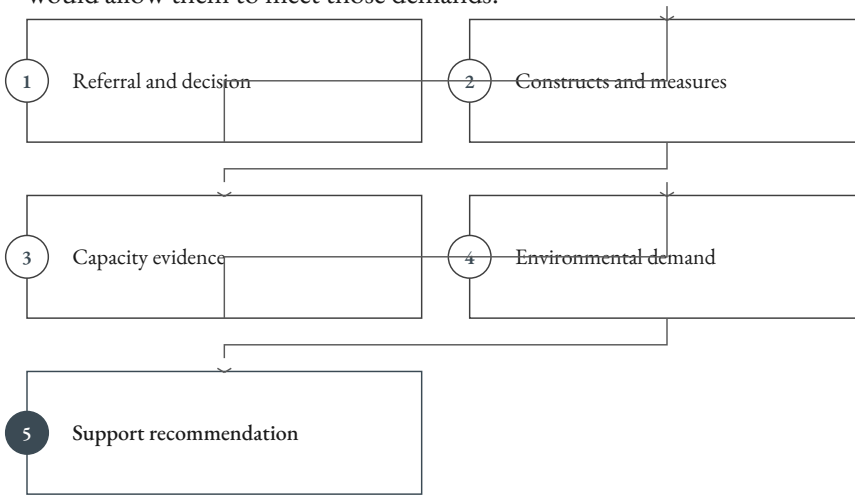


Figure 7.1. Functional assessment links capacity, demand and support. The decision concerns the fit between them, not the capacity alone.

7.1 Academic Achievement Assessment

Achievement assessment measures what a person has learned in specified domains relative to a reference group or a curriculum. Its purpose is usually

to identify a learning difficulty, to establish a baseline for intervention, or to evaluate whether an intervention has worked.

Achievement and ability are different constructs

Achievement measures developed skill; ability measures developed capacity. The distinction matters because the two are highly correlated but not interchangeable, and because some measures described as achievement tests are in fact measures of reasoning applied to academic content. The assessor should know which construct a given instrument measures, since the interpretation differs: a low achievement score with average ability suggests a skill-specific difficulty, whereas a low achievement score with low ability suggests a general learning difficulty requiring a different response.

Curriculum-based and norm-referenced approaches

Norm-referenced achievement tests compare a student with a national or regional sample. They are useful for identifying significant under-attainment and for establishing eligibility for services, and they are insensitive to what a particular student has actually been taught. Curriculum-based measurement compares performance with expected progress through a specified curriculum and is more sensitive to short-term change and to instructional effects. The two approaches answer different questions and are complementary; an assessment that uses only one will have a blind spot.

Assessing the conditions of learning

Achievement scores describe performance under the conditions of testing. The assessor should record the conditions of learning that produced that performance: attendance, language of instruction, teaching quality, class size, disruption, vision and hearing, and the student's own account of what they find difficult. A student who has missed a third of the school year has an attendance problem that no test can diagnose, and an assessment that fails to record this has produced a number without a meaning.

7.2 Reading, Writing and Mathematics

Each of the three core academic domains has a structure that determines what must be assessed and how findings translate into instruction.

Reading

Reading comprises word-level skills (phonological awareness, decoding, sight-word recognition, fluency) and comprehension (vocabulary, inference, background knowledge, working memory). Assessment must establish which component is impaired, because the interventions differ: a decoding difficulty requires systematic phonics instruction, whereas a comprehension difficulty in a fluent decoder requires vocabulary and strategy work. A composite reading score conceals the distinction, and reporting only a composite score is a common and consequential error.

Writing

Writing involves transcription (handwriting, spelling), text generation (idea production, organisation) and executive control (planning, revising). Difficulties at the transcription level consume resources that would otherwise serve composition, so a student with slow handwriting or poor spelling may produce weak written work despite adequate ideas. Assessment should therefore separate transcription from composition, and should include at least one task that allows the student to dictate or use a keyboard where that is the accommodation under consideration.

Mathematics

Mathematics comprises number sense, procedural fluency, fact retrieval, problem representation and reasoning. Difficulties in fact retrieval and in procedural fluency are distinguishable from difficulties in mathematical reasoning, and the distinction predicts response to intervention. Assessment should include untimed reasoning tasks as well as timed calculation, because a student whose reasoning is intact but whose retrieval is slow requires a different accommodation from one whose reasoning is impaired.

Table 7.1. Academic domains, component skills and assessment implications

Domain	Components to assess	Distinguishing finding	Instructional implication
Reading — word level	**Phonological awareness; decoding; fluency**	**Accurate but slow decoding; poor nonword reading**	**Systematic phonics; fluency practice**
Reading — comprehension	**Vocabulary; inference; background knowledge**	**Fluent decoding with weak comprehension**	**Vocabulary and strategy instruction**
Writing — transcription	**Handwriting fluency; spelling**	**Slow, effortful handwriting; poor spelling**	**Transcription intervention; keyboard access**
Writing — composition	**Planning; text generation; revision**	**Adequate transcription, weak organisation**	**Strategy instruction; planning scaffolds**
Mathematics — fluency	**Fact retrieval; procedural accuracy**	**Accurate but slow and effortful calculation**	**Timed practice; calculator access where appropriate**
Mathematics — reasoning	**Problem representation; multi-step reasoning**	**Adequate fluency, weak word-problem performance**	**Explicit reasoning instruction; language support**
Cross-domain	Attention; working memory; language proficiency	Broadly depressed performance in one language only	Language assessment before diagnosis

7.3 Aptitude and Learning Potential

Aptitude measures predict the capacity to acquire skills under specified conditions of training or instruction. They are used in educational selection, in occupational settings and, in counselling psychology, in the assessment of learning potential for people whose educational history does not permit a straightforward interpretation of achievement scores.

Aptitude, ability and achievement

The boundaries between the three constructs are partly conventional. A test of verbal reasoning used to predict academic success is an aptitude measure; the same test used to estimate current cognitive capacity is an ability measure. What distinguishes aptitude is the predictive purpose: the score is interpreted as a forecast rather than a description. Forecasts are probabilistic and depend on the criterion, the training conditions and the range of abilities in the group to which the forecast applies.

Dynamic assessment and learning potential

Dynamic assessment measures the degree to which a person benefits from structured mediation, usually by comparing performance before and after a teaching phase. Its rationale is that learning potential is a better predictor of response to instruction than current attainment, particularly for people whose prior opportunities to learn have been limited. The approach has a substantial literature and a modest evidence base for incremental prediction; its principal value in counselling psychology is in assessment of people with atypical educational histories, where static scores under-estimate capacity.

Cautions

Aptitude scores should not be used to foreclose opportunity. The prediction of a group's average performance is a poor basis for a decision about an individual, and the lower the score, the less reliable the prediction of what that individual can achieve given adequate support. Where aptitude scores are used in selection, the assessor should know the selection utility research for the specific instrument and setting, and should recommend the use of multiple predictors.

7.4 Career Interests and Values

Career assessment addresses the fit between a person's interests, values, abilities and the demands and rewards of occupations. It is used in career counselling, in vocational rehabilitation, in educational guidance and in occupational settings where career transitions are being planned.

Interest inventories

Interest inventories, most of which are built on Holland's (1997) RIASEC model of realistic, investigative, artistic, social, enterprising and conventional themes, measure the similarity between a person's pattern of interests and the patterns characteristic of occupational groups. Their predictive validity for occupational choice and satisfaction is moderate and well established. Their limitations are equally established: interests are shaped by opportunity and by gender and class socialisation; they describe preferences, not capacities; and they can be constrained by what a person knows about the world of work.

Work values and motivators

Values inventories assess what a person wants from work: security, autonomy, recognition, service, income, work–life balance, intellectual challenge. Values predict satisfaction and turnover better than interests predict performance, and they are particularly useful in mid-career transitions where the person's priorities have changed. The assessor should treat values as context-dependent and revisable rather than as fixed traits.

Interpreting career data

Career assessment is most useful when it broadens rather than narrows the client's options. Interpreting a profile as a prescription — "your profile indicates accountancy" — is a misuse. The appropriate interpretation identifies themes, explores how they have been expressed, considers the constraints of the local labour market, and generates options for exploration. The client's own knowledge of their circumstances should carry substantial weight.

7.5 Work Personality and Work Adjustment

Beyond interests and values, certain personal characteristics predict adjustment to work: conscientiousness, emotional stability, interpersonal style, tolerance for stress and capacity for co-operation. These are the substance of what is sometimes called work personality.

Assessment approaches

Work personality is assessed through broad personality inventories, through specialised measures of work-related behaviour, through interviews about work history, and through structured references. Work history is often the most informative source and the most neglected: a pattern of short tenures, repeated conflicts with supervisors, or rapid disengagement after promotion carries more predictive weight than a trait score, provided the pattern is examined for situational explanations.

Work adjustment and its determinants

Work adjustment is a function of the person, the job demands, the supervisory relationship, the organisational climate and the fit between them. Assessment that characterises the person without characterising the job is incomplete. Where possible, the assessor should obtain a description of the role's demands, the physical and interpersonal environment, and any changes that preceded the difficulty; occupational health services, job descriptions and supervisor accounts are the usual sources.

Reporting work-related findings

Reports for occupational audiences should be behavioural and practical. Statements about personality traits are of limited use to a manager; statements about the conditions under which the person performs effectively, and the accommodations that would support performance, are of direct use. This is also the form of reporting least likely to stigmatise.

7.6 Adaptive Behaviour

Adaptive behaviour describes the conceptual, social and practical skills required for everyday functioning. It is central to the diagnosis of intellectual disability, where significant limitations in adaptive functioning are a defining requirement, and it is relevant wherever functional capacity is at issue.

Conceptual, social and practical domains

Conceptual skills include literacy, numeracy, reasoning and self-direction. Social skills include interpersonal communication, social judgement, and the capacity to follow rules and avoid victimisation. Practical skills include personal care, domestic tasks, use of money, use of transport and occupational functioning. Assessment requires informant report, because self-report in this area is systematically unreliable, and it requires attention to the environments in which the person actually functions.

Instruments and methods

Standardised adaptive behaviour instruments, such as the Vineland Adaptive Behavior Scales (Sparrow, Cicchetti, & Saulnier, 2016) and the Adaptive Behavior Assessment System, are completed by an informant who knows the person well and yield standard scores by domain. Direct observation and task-based assessment supplement informant report, and are particularly useful where informants disagree or where the person lives in a highly structured setting that may mask or compensate for limitations.

Distinguishing adaptive behaviour from ability

Adaptive behaviour and intellectual ability are correlated but distinct, and the discrepancy between them is clinically important. A person with mild intellectual disability may function independently with support; a person with average ability may have severe adaptive limitations due to autism, mental illness or institutionalisation. Diagnosis of intellectual disability requires both, and assessment of only one is a recurring source of error.

7.7 Functional Capacity and Daily Activities

Functional capacity assessment evaluates what a person can do in the activities of daily living, usually for the purpose of a decision about support, accommodation or eligibility. It differs from adaptive behaviour assessment in its focus on specific tasks and specific environments rather than on developmental skill level.

Defining the activity

The first task is to define the activity precisely. "Can he manage his finances?" is not assessable; "can he pay household bills on time, keep track of a bank balance and avoid fraudulent transactions?" is. Domains commonly assessed include self-care, mobility, meal preparation, household management, medication management, financial management, communication, transport, social participation and work-related tasks.

Methods

Functional capacity is assessed through interview, informant report, direct observation and, where possible, task performance in a realistic environment. Standardised functional measures and structured checklists improve reliability; occupational therapy assessment is often the most thorough source. Self-report is the weakest method for this purpose, since people systematically over-estimate or under-estimate their capacity, sometimes for understandable reasons.

The capacity–performance distinction

A person may have the capacity to perform an activity but not perform it in practice, because of environmental barriers, mental state, motivation or lack of opportunity. Conversely, a person may perform an activity only because of extensive support that is invisible in the assessment. Reports should distinguish capacity, performance and support, because decisions about support depend on all three.

7.8 Disability and Accommodation Decisions

Psychological evidence frequently contributes to decisions about disability status, reasonable accommodations and eligibility for services. These decisions are legal and administrative, not clinical, and the assessor's role is to supply evidence relevant to the criteria rather than to make the determination.

Knowing the framework

Each jurisdiction and institution defines disability and accommodation differently. The assessor should know the relevant definitions, the evidence required, the standard of proof and the decision-maker's criteria. Producing a clinical report that does not address the criteria is a common failure, and it is avoidable by reading the guidance before conducting the assessment.

The accommodation question

Accommodation decisions turn on three questions: whether the person has a functional limitation relative to the demands of the setting; whether an adjustment would remove or reduce the barrier; and whether the adjustment would fundamentally alter the nature of the activity or impose disproportionate cost. The assessor can contribute evidence on the first two and should be cautious about the third, which is usually a legal or managerial judgement.

Evidence quality

Where the decision has significant consequences, the evidence should be convergent: standardised measures, functional observation, history and, where relevant, medical documentation. Single-measure evidence is weak, and the assessor should say so rather than overstate confidence. Where the evidence is genuinely ambiguous, the report should present the ambiguity rather than resolve it in whichever direction seems kinder.

7.9 Career Assessment with Adults in Transition

Career transitions — redundancy, injury, migration, caring responsibilities, mid-life reassessment — are among the most common reasons for functional and vocational assessment in counselling psychology. They require an approach that integrates capacity, interest, values, context and opportunity.

The assessment frame

Transition assessment begins with the person's own account of the change and its meaning. A forced exit from a long-held role involves loss and identity disruption that will affect how any assessment is received. The assessor should establish what the person wants, what constraints they face, what resources they have and what they have already tried, before introducing standardised measures.

Integrating data

The integration combines three sources: the person's history of what they have done and enjoyed; standardised measures of interest, values and aptitude; and information about the labour market and training opportunities available to them. Where these conflict, the conflict is the material for counselling rather than for a verdict. A person whose interests point towards work their health cannot support requires a conversation about adaptation, not a recommendation that ignores one side of the evidence.

Realistic optimism

The assessor's responsibility is to be realistic without being discouraging. Evidence about prognosis, labour market conditions and the effects of support should be presented honestly, together with the evidence that targeted support substantially improves return-to-work outcomes for many groups. A report that presents only constraints is as unhelpful as one that presents only encouragement.

7.10 Integrating Educational and Career Data

Where an assessment covers both educational and occupational domains, integration is required to prevent contradictory recommendations. The most common contradiction arises where academic recommendations assume a level of support that the occupational recommendations do not, or where career recommendations assume a capacity that the cognitive findings do not support.

A coherence check

Before finalising such a report, the assessor should check that the recommendations are consistent with each other, consistent with the evidence, and feasible in the settings concerned. Where a recommendation depends on an assumption — that a school will provide a reader, that an employer will permit flexible hours — the assumption should be stated so that the reader can judge whether it holds.

Involving the person and the setting

Integration is improved by involving the client and, where appropriate, the receiving setting in the planning of recommendations. A recommendation developed with a school's special educational needs coordinator, or with an employer's occupational health adviser, is more likely to be implemented than one written in isolation. This is not a compromise of independence; it is an application of the principle that assessment should be useful.

7.11 Ethical Use of Occupational Tests

Occupational testing raises ethical issues that are less prominent in clinical assessment because the test-taker is often not the client, the results affect selection, and the assessor may be retained by the organisation.

Consent, feedback and access

Candidates have a right to know what is being assessed, to receive feedback in a form they can understand, and to have their data handled confidentially. These rights are frequently poorly met in selection contexts, where tests are administered at scale and results are reported only to the employer. The psychologist's obligations do not disappear because the organisation is the client; they require negotiation at the outset.

Fairness and adverse impact

Selection tests that produce substantially different pass rates for different groups require justification on grounds of job relevance. The assessor should know whether the instrument has evidence of validity for the

specific role and whether adverse impact has been examined, and should advise against instruments and cut scores that have not been justified. This is both an ethical and a legal requirement in many jurisdictions.

Prediction, not prophecy

Finally, occupational tests predict probabilities, not destinies. A test score that predicts a 60 per cent chance of satisfactory performance says nothing definitive about an individual, and using it as a cut-off has consequences that the assessor should be prepared to defend. Where the assessor is asked to make an individual decision from group data, the appropriate response is to state the limits of the prediction and to recommend the use of additional evidence.

7.12 Writing Functional Recommendations

Functional recommendations are the practical output of this family of assessments, and their quality determines whether the assessment has any effect. Table 7.2 contrasts weak and strong recommendations across common domains.

Table 7.2. Weak and strong functional recommendations

Domain	Weak formulation	Strong formulation
Examinations	"Needs extra time"	"Twenty-five per cent additional time in written examinations, reviewed after one term against timed writing samples"
Reading	"Struggles with reading"	"Access to audiobooks for set texts; phonics-based intervention three times weekly for twelve weeks"
Medication	"Needs help with medication"	"Weekly blister pack prepared by a pharmacist; daily check by a support worker; review in eight weeks"
Workload	"Reduce stress at work"	"Maximum of two concurrent projects; written specifications for each; fortnightly supervision for the first three months"
Transport	"Unable to travel independently"	"Door-to-door transport for appointments; travel training assessment after three months"
Review	"Review as needed"	"Review after twelve weeks, with repeat of the timed reading measure and feedback from the line manager"

Principles

A strong recommendation names the action, the agent, the setting and the review point. It distinguishes what is required from what is desirable. It is proportionate to the evidence, and it does not smuggle in clinical claims that the assessment did not establish. It is written in language the implementing party can act on without further translation.

Implementation and review

Recommendations should be reviewed. A recommendation that has not been implemented is not evidence that the assessment was wrong, but it is information that the assessor should seek, because the reasons for non-implementation — cost, feasibility, misunderstanding — are usually remediable. Follow-up of a sample of recommendations is one of the most effective ways for a service to improve its reports.

Advanced Case Conference

The referral. A vocational rehabilitation service asks for assessment of a 41-year-old man who sustained a moderate traumatic brain injury two years ago in a road traffic collision. He wishes to return to work as a site electrician. The question is what support he would need and whether a return is realistic.

The assessment. Cognitive assessment shows high-average verbal ability, average perceptual reasoning, low-average working memory and borderline processing speed. Memory is average on recognition but weak on free recall, improving with structure. Executive measures show difficulty with planning and with inhibiting responses under time pressure. Fatigue is marked: performance declines across the session, and he reports needing two hours of rest each afternoon. He has not worked since the injury and has completed a part-time retraining course with support. Performance validity is adequate.

The reasoning. His technical knowledge is preserved and his recognition memory is adequate, so a return to work involving familiar procedures with written reference materials is plausible. The limiting factors are processing speed under time pressure, planning under novel conditions, and fatigue, all of which are hazardous on a construction site. A return to the full role without adjustment would place him and others at risk; a return to a modified role with supervision, written procedures and reduced hours is consistent with the evidence.

The decision. The report recommends a phased return over sixteen weeks, with a maximum six-hour day initially, no work at height or on live equipment until reviewed, written job specifications and a named mentor. It recommends that a risk assessment be conducted jointly with the employer, and that fatigue be monitored with a simple daily log. It states that the prognosis for a return to the full role is uncertain, and that the review at sixteen weeks should determine whether the modification becomes permanent.

The teaching point. The assessment's value lay in the specificity of the recommendation and in its explicit engagement with risk. A report that had said "fit for light duties" would have been unusable, and possibly dangerous.

Chapter Summary

Achievement, aptitude, adaptive behaviour and functional capacity are distinct constructs, and the assessor must match the instrument to the

construct the decision requires. Academic assessment should identify the component skill that is impaired, because the instructional implication differs by component. Interest and values inventories are useful for broadening career exploration and should not be used prescriptively. Adaptive behaviour requires informant report and is conceptually distinct from ability. Functional capacity assessment must define the activity, distinguish capacity from performance and support, and engage with the legal or administrative criteria that frame the decision. Recommendations are the practical output of this work, and their quality depends on specificity, proportionality and a named review point.

Review Questions

1. Distinguish achievement, aptitude and ability, and explain why the distinction affects interpretation.
2. Why should reading assessment separate decoding from comprehension?
3. What does dynamic assessment add to static testing, and what are its limits?
4. Why is self-report a weak method for assessing adaptive behaviour?
5. Identify the four elements of a strong functional recommendation.

Discussion Questions

1. A school asks you to assess a student to support an application for additional funding. How do you manage the potential conflict between the school's interest and the child's?
2. Should interest inventories be used to steer young people away from occupations? Defend your position.
3. How would you assess functional capacity for a client who lives alone and has no informant?
4. What are the ethical obligations of a psychologist retained by an employer to test candidates?

5. How much weight should an assessor give to a client's stated career goal that the evidence suggests is unrealistic?

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PART III

Assessing Clinical Presentations

Mood, anxiety, obsessive-compulsive, trauma-related, bipolar, psychotic, substance-related and enduring personality presentations.

Chapter 8 · Chapter 9 · Chapter 10 · Chapter 11

CHAPTER 8

Assessment of Mood, Anxiety, Obsessive-Compulsive, Trauma and Stress-Related Problems

Learning Objectives

After working through this chapter, the reader should be able to:

- structure the assessment of depressive and anxiety presentations around differential reasoning;
- distinguish panic, social anxiety, generalised anxiety and obsessive-compulsive presentations;
- assess trauma exposure and post-traumatic symptomatology without assuming aetiology;
- appraise the boundary between adjustment reactions and clinical disorder;
- select and interpret severity measures for monitoring rather than labelling;
- conduct a functional analysis of avoidance that informs treatment planning.

Key terms: internalising spectrum · anhedonia · panic attack · agoraphobic avoidance · obsession · compulsion · intrusion · hyperarousal · adjustment disorder · comorbidity · functional analysis · severity monitoring · reliable change

Introduction

The internalising disorders — depression, the anxiety disorders, obsessive-compulsive disorder and the trauma- and stressor-related disorders — account for a large share of presentations in counselling psychology services. They are also the group for which the evidence base for assessment is strongest: there are well-validated diagnostic interviews,

severity measures with published norms, and a substantial literature on measurement-based care.

That strength carries a risk. Because these conditions are familiar, assessment is often conducted by pattern recognition, and because validated questionnaires are widely available, scores are often used as diagnoses. This chapter argues for a more disciplined approach: structured differential reasoning that considers the neighbouring conditions, careful attention to the boundary between disorder and ordinary distress, and the use of severity measures for monitoring and comparison rather than for labelling.

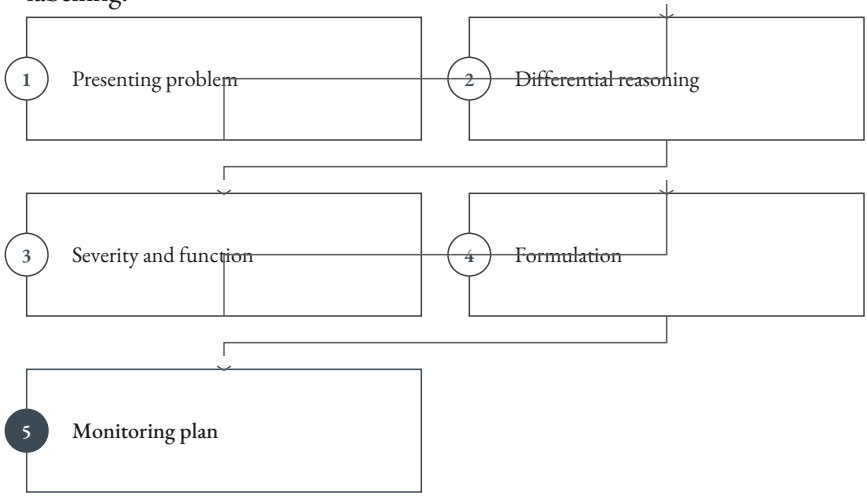


Figure 8.1. Assessment across the internalising spectrum. Overlapping presentations require explicit discrimination rather than reliance on the most salient symptom.

8.1 Assessing Depressive Presentations

Depression is the most common presentation in most adult mental health services and the one most often assessed inadequately. The core features — persistent low mood and loss of interest or pleasure — are widely known; what distinguishes a competent assessment is attention to course, severity, subtype, differential and risk.

Core features and their assessment

Low mood must be distinguished from the reactive sadness of ordinary disappointment, and anhedonia from the temporary withdrawal that follows loss. The assessor establishes pervasiveness across contexts, persistence across the day, and the presence of biological features: early waking, appetite change, psychomotor change and diurnal variation. These features have diagnostic and prognostic significance: early waking with diurnal worsening of mood suggests a melancholic picture, which responds differently to treatment from a reactive depression with preserved reactivity.

Course and subtype

Course questions establish whether the presentation is a first episode, a recurrence or a chronic state. The distinction between a major depressive episode superimposed on persistent depressive disorder and a first episode matters for prognosis and for the length of treatment recommended. Subtype questions — melancholic, atypical with hypersomnia and leaden paralysis, psychotic, seasonal, peripartum, catatonic — carry implications for treatment choice and for risk.

Severity and functional impact

Severity should be established by symptom count, functional impact and risk, not by a questionnaire score alone. A client with moderate symptoms who has stopped working, is neglecting self-care and has begun drinking heavily has a severe presentation whatever the score suggests. Severity measures are useful as a baseline for monitoring and as a check on clinical impression; they are not a substitute for either.

Differential and comorbidity

The differential includes bipolar disorder, substance-induced mood disorder, adjustment disorder, grief, thyroid disease and anaemia, medication effects, and the negative symptoms of schizophrenia. Bipolar disorder is the most consequential item to miss, and missing it can lead to

harmful treatment. Any history of elevated mood, decreased need for sleep, increased goal-directed activity or a family history of bipolar disorder requires specific enquiry, ideally with informant corroboration and prospective mood charting.

8.2 Assessing Anxiety Presentations

Anxiety disorders are heterogeneous, frequently comorbid with each other, and easily confused in unstructured assessment. The assessor's task is to identify the specific fear structure rather than to record a general elevation of anxiety.

The specific phobic and anxiety presentations

Panic disorder is characterised by recurrent unexpected panic attacks and persistent concern about further attacks or their consequences. Agoraphobia is characterised by fear of situations from which escape might be difficult or help unavailable. Social anxiety disorder involves fear of negative evaluation in social or performance situations. Generalised anxiety disorder involves excessive, difficult-to-control worry across multiple domains, accompanied by arousal symptoms. Separation anxiety disorder and selective mutism are relevant in younger populations.

Establishing the fear structure

For each presentation, the assessor establishes what is feared, what triggers it, what the person does to prevent or escape it, and what the consequences are. This information is more useful than a severity score, because it determines the treatment target. A client who fears the physical sensations of panic requires interoceptive work; a client who fears negative evaluation requires behavioural experiments in social situations; a client who fears that worry prevents disaster requires metacognitive work.

Generalised anxiety and its boundary with worry

Generalised anxiety disorder is the most difficult of the anxiety disorders to assess reliably, because worry is a universal experience and the boundary

between normal and pathological worry is quantitative rather than qualitative. The discriminating features are uncontrollability, pervasiveness across domains, duration of at least six months, and associated symptoms. The assessor should also consider whether the worry is secondary to another condition, such as health anxiety, obsessive-compulsive disorder or a depressive episode with ruminative thinking.

8.3 Panic and Agoraphobic Symptoms

Panic attacks are discrete episodes of intense fear accompanied by somatic and cognitive symptoms that peak within minutes. They are a feature of several disorders and are not a diagnosis in themselves; the diagnostic question concerns the pattern and the person's response to it.

Assessing the attack

A careful description establishes the symptoms present, their speed of onset, their peak intensity, their duration and the person's interpretation of them. The interpretation is central: a person who believes that palpitations signify a heart attack has a different maintaining mechanism from one who believes they are going to faint. The assessor should also establish whether attacks are unexpected, situationally bound or situationally predisposed, since this distinction shapes diagnosis.

Panic disorder with and without agoraphobia

The distinction between panic disorder and agoraphobia is not merely a matter of symptom count. Agoraphobic avoidance can occur without panic, and panic can occur without avoidance. Where avoidance is present, the assessor should map the avoided situations, the safety behaviours used, and the beliefs that sustain them. This map is the treatment plan in embryo.

Medical differential

Panic symptoms overlap with cardiac, respiratory, endocrine and neurological conditions, and with the effects of caffeine, stimulants and

withdrawal states. A first presentation of panic in later life, panic with atypical features, or panic unresponsive to treatment warrants medical review. The assessor should record what medical evaluation has occurred and what remains outstanding.

8.4 Obsessions and Compulsions

Obsessive-compulsive disorder is defined by obsessions (intrusive, unwanted thoughts, images or urges) and compulsions (repetitive behaviours or mental acts performed to reduce distress or prevent a feared outcome). It is frequently misdiagnosed as generalised anxiety, as depression, or as a personality characteristic such as perfectionism.

Distinguishing obsessions from ordinary intrusive thoughts

Intrusive thoughts are near-universal; what distinguishes clinical obsessions is their frequency, their appraisal as significant and threatening, the distress they cause, and the effort expended to neutralise them. The assessor should ask whether the person believes the thought is meaningful about them (thought–action fusion), whether they attempt to suppress or neutralise it, and how much time it consumes.

Assessing compulsions, including mental rituals

Compulsions are not always visible. Mental rituals — counting, praying, reviewing, replacing a thought with a safe thought, mentally checking — are common and easily missed. The assessor should ask directly about covert rituals and should ask about avoidance, since avoidance is functionally equivalent to a compulsion and often more disabling.

Subtypes, insight and related presentations

Subtypes include contamination, symmetry and ordering, forbidden or taboo thoughts, and hoarding, which is classified separately. The assessor should record the degree of insight: many people with OCD recognise that their fears are excessive, and a minority do not, which affects both diagnosis and engagement. Body dysmorphic disorder, trichotillomania, skin-picking

disorder and olfactory reference syndrome are related presentations that should be screened where relevant.

8.5 Trauma Exposure and Post-Traumatic Symptoms

Trauma-related assessment requires attention to what happened, how the person has responded, and what the current symptom profile is. It also requires a discipline that is often absent: distinguishing the presence of trauma from the presence of post-traumatic stress disorder, since most people who experience trauma do not develop the disorder.

Establishing exposure

Exposure is established by asking about specific categories of event — serious accidents, physical or sexual assault, combat, natural disaster, sudden bereavement, childhood abuse, witnessing violence, and events experienced in a professional capacity — with attention to the person's own definition of what was traumatic. The assessor should record the nature of the event, its duration, the person's age at the time, the degree of threat to life or bodily integrity, and whether the event was interpersonal, since interpersonal trauma carries different prognostic and relational implications.

The symptom clusters

DSM-5-TR and ICD-11 organise post-traumatic symptoms differently. The DSM-5-TR clusters are intrusion, avoidance, negative alterations in cognition and mood, and alterations in arousal and reactivity, with a dissociative subtype; ICD-11 uses a narrower set of core symptoms (re-experiencing, avoidance and a sense of current threat) with optional disturbances in affect regulation, self-concept and relationships. The assessor should know which framework is being applied, because the two produce different thresholds for the same presentation.

Differential and comorbidity

The differential includes acute stress disorder, adjustment disorder, complex post-traumatic stress disorder, dissociative disorders, borderline personality disorder, psychotic disorders with trauma-related content, and traumatic brain injury. Comorbidity with depression, substance use and chronic pain is the rule. Assessment should establish which condition is primary for treatment purposes, rather than assuming that all symptoms are trauma-related because a trauma is present.

8.6 Stress and Adjustment Problems

Adjustment disorder describes a maladaptive response to an identifiable stressor, with symptoms that do not meet criteria for another disorder, beginning within three months of the stressor and resolving within six months of its termination. Its inclusion in both DSM-5-TR and ICD-11 has been controversial, and its boundaries are genuinely uncertain.

The clinical problem

The central difficulty is that adjustment disorder is defined by exclusion: it applies when symptoms are clinically significant but do not meet criteria for another condition. This makes it both common and vulnerable to over-use, particularly in services where a diagnosis is required for access and the presentation is subthreshold. The assessor should therefore be explicit about which criteria are met and which conditions have been excluded, rather than using adjustment disorder as a default.

Distinguishing disorder from distress

The distinction between an understandable reaction and a disorder rests on severity, functional impairment, duration and the presence of symptoms beyond those expected for the stressor. Grief is the paradigm case: the DSM-5-TR removed the former bereavement exclusion from major depression, while retaining a category of prolonged grief disorder for reactions that persist and impair. The assessor should not pathologise normal grief, and should not dismiss persistent, impairing grief as normal.

Assessment tasks

Assessment establishes the nature and timing of the stressor, the person's appraisal of it, the symptoms, the functional impact, the available supports, and the person's own understanding. It should also establish what the person has already tried and what they believe would help, since adjustment difficulties often respond to brief, practical intervention.

8.7 Somatic Overlap and Medical Explanations

Many presentations in this chapter include prominent somatic symptoms, and some are better explained by a medical condition. The assessor's obligation is to consider medical explanations, to record what has been excluded, and to avoid attributing to psychology what has not been medically evaluated.

Somatic presentations

Somatic symptom disorder, illness anxiety disorder, functional neurological symptoms and pain syndromes overlap substantially with anxiety and depression. The assessor should establish the history of medical investigation, the person's beliefs about their symptoms, the degree of functional impairment, and the pattern of health-care use. Where symptoms are disproportionate to identifiable disease, the assessment should address the psychological maintaining factors without implying that the symptoms are imaginary.

Medical mimics and contributors

Thyroid disease, anaemia, vitamin deficiency, sleep apnoea, chronic pain, medication effects, substance withdrawal and neurological conditions can all present with depressive or anxiety symptoms. The assessor should ask about recent investigations and should recommend medical review where symptoms are atypical, refractory or associated with physical signs. A collaborative relationship with the referring physician is the most efficient safeguard.

8.8 Severity Measures and Symptom Monitoring

Standardised severity measures improve the reliability of monitoring and support measurement-based care, in which treatment decisions are informed by repeated measurement rather than by clinical impression alone. Their value depends on using them correctly.

Selecting a measure

The instrument should match the construct, the population and the purpose. Widely used examples include the Patient Health Questionnaire-9 for depression (Kroenke, Spitzer, & Williams, 2001), the Generalised Anxiety Disorder-7 for anxiety (Spitzer, Kroenke, Williams, & Löwe, 2006), the Yale–Brown Obsessive Compulsive Scale for OCD, and the PTSD Checklist for DSM-5. Each has published norms, cut scores and, in some cases, reliable change indices. The assessor should know the instrument's accuracy in the population being assessed, which frequently differs from the population in which it was validated.

Using measures for monitoring

For monitoring, the relevant properties are sensitivity to change, test–retest reliability in stable clients, and a reliable change index. A change that exceeds the reliable change index is unlikely to be measurement error; a change that does not may still be clinically meaningful, and the assessor should combine the score with functional information. Weekly or fortnightly administration is typical in treatment monitoring; the same instrument should be used throughout to permit comparison.

Avoiding the score-as-diagnosis error

A severity score is not a diagnosis. Scores above a cut-off indicate probable caseness and warrant diagnostic assessment; scores below it do not exclude significant difficulty, particularly where the instrument does not cover the person's presentation. The assessor should report scores with their confidence intervals, their percentile or severity band, and their date, and should never present a score as the basis for a diagnosis without a diagnostic

interview.

8.9 Functional Analysis of Avoidance

Avoidance is the most important maintaining process across the internalising disorders. A functional analysis of avoidance identifies the situations avoided, the behaviours used to avoid or escape them, the short-term consequences that reinforce the avoidance, and the long-term costs that sustain the disorder.

The analysis

The analysis proceeds through four questions. What is the feared outcome? In which situations is it expected? What does the person do to prevent or escape it? What happens afterwards? The answers yield a formulation in which the avoidance is not a symptom to be removed but a solution that has become the problem: it reduces anxiety immediately and prevents disconfirmation in the long run.

Safety behaviours and subtle avoidance

Avoidance includes not only overt non-participation but subtle safety behaviours: gripping a handrail, carrying medication, rehearsing sentences, sitting near the exit, drinking alcohol before social events. These behaviours are often not reported unless specifically asked about, because they are experienced as sensible precautions rather than as avoidance. Their identification is essential, since treatment requires them to be dropped.

From analysis to plan

The functional analysis translates directly into a treatment plan: graded exposure to the avoided situations, elimination of safety behaviours, behavioural experiments designed to test the feared outcome, and monitoring of the predictions and their outcomes. Where the analysis cannot be conducted — because the client is unable to describe the pattern, or because the presentation is too acute — the assessment should record that and specify what needs to be established before treatment planning

can proceed.

8.10 Differential Diagnosis Across Internalising Problems

The internalising disorders overlap substantially in symptoms, in comorbidity and in underlying temperament. Distinguishing among them requires attention to the specific content of fear and to the temporal pattern, rather than to symptom counts.

The common differentials

The most frequently confused pairs are generalised anxiety and major depression (both characterised by worry, sleep disturbance and fatigue); panic disorder and generalised anxiety (both involve arousal symptoms); social anxiety and avoidant personality disorder (both involve social fear and avoidance); obsessive-compulsive disorder and generalised anxiety (both involve repetitive worry); and post-traumatic stress disorder and adjustment disorder (both follow stressors).

Discriminating features

The discriminating features are usually specific: the presence of unexpected panic attacks; the presence of obsessions and compulsions; the presence of re-experiencing and avoidance of trauma reminders; the presence of a stable, lifelong pattern of social fear beginning in adolescence; the temporal relationship to a stressor. Table 8.1 sets out the principal distinctions.

Table 8.1. Discriminating features among commonly confused internalising presentations

Presentation	Distinguishing feature	Frequent confusions
Major depressive episode	Pervasive anhedonia; biological features; episode duration of at least two weeks	Grief; bipolar depression; adjustment disorder
Generalised anxiety disorder	Uncontrollable worry across multiple domains for at least six months	Depression with rumination; OCD; health anxiety
Panic disorder	Recurrent unexpected attacks with persistent concern about further attacks	Medical conditions; PTSD hyperarousal; social anxiety
Social anxiety disorder	Fear of negative evaluation in social or performance situations; early onset	Avoidant personality disorder; agoraphobia; autism
Obsessive-compulsive disorder	Obsessions with attempts to neutralise; covert and overt rituals; time consumed	Generalised anxiety; perfectionism; psychotic phenomena
Post-traumatic stress disorder	Re-experiencing, avoidance and current-threat symptoms after a qualifying event	Adjustment disorder; complex PTSD; dissociative disorders
Adjustment disorder	Symptoms within three months of an identifiable stressor, not meeting another disorder	Subthreshold depression or anxiety; normal stress response

Comorbidity and hierarchy

Where criteria for several disorders are met, the assessor should record all of them rather than choosing one, and should identify which is primary for treatment. Diagnostic hierarchies — rules that give one disorder precedence — have been largely removed from the current classifications because they obscured clinically important comorbidity, and their removal requires the assessor to reason about priority explicitly.

8.11 Comorbidity and Diagnostic Hierarchy

Comorbidity is the norm in this group. Depression with anxiety, anxiety with substance use, OCD with depression, and PTSD with all of the above are common patterns rather than exceptions. The assessor's task is not to reduce the diagnoses to one but to establish which condition should be treated first and why.

Reasons for comorbidity

Comorbidity arises from shared risk factors (neuroticism, childhood adversity, genetic liability), from causal relationships (anxiety leading to depression, trauma leading to substance use), from overlapping diagnostic criteria, and from help-seeking patterns that bring comorbid people into services. Distinguishing these mechanisms matters for treatment: where one condition causes another, treating the cause may resolve both; where they are independent, both may require treatment.

Establishing priority

Priority is established by severity, by functional impact, by the client's own goals, by the presence of risk, and by the evidence on sequencing. Where depression is severe and anxiety moderate, treating depression first is usually appropriate; where panic is preventing engagement in treatment, treating panic first may be necessary. The reasoning should be recorded, because the next clinician will need to know why the order was chosen.

The single-diagnosis temptation

Services that require a single diagnosis for access create pressure to record only one condition. The assessor should resist this where the evidence supports comorbidity, and should record the additional diagnoses in a way that the service can accommodate — for example, as secondary diagnoses or as clinical formulation. Deliberately omitting a diagnosis to satisfy an administrative category is a misrepresentation of the evidence.

8.12 Integrated Formulation and Feedback

The output of assessment in this group is a formulation that links the presentation to its development, its maintenance and its treatment. It should explain why this person, at this time, has developed this particular pattern, and it should identify the points at which intervention is most likely to be effective.

Structure of the formulation

A workable structure moves through predisposing, precipitating, perpetuating and protective factors, and then specifies the maintaining mechanisms. The mechanisms should be stated in a form that predicts treatment response: if avoidance is the principal maintaining process, exposure should reduce symptoms; if rumination is maintaining depression, rumination-focused work should help.

Feedback

Feedback should present the formulation as a shared account rather than a verdict, use the client's own language where possible, invite correction, and connect the formulation to the treatment options. Where the assessment has identified a condition the client did not expect — bipolar disorder, OCD, PTSD — the feedback should include information about the condition, its treatability and the rationale for the proposed approach. Where it has excluded a condition the client expected, the exclusion should be explained rather than simply recorded.

Monitoring and review

The assessment should end with a monitoring plan: which measures will be repeated, at what intervals, and what change would prompt a revision of the formulation or the treatment. Measurement-based care is only as good as the arrangement that sustains it, and the assessment report is the natural place to establish that arrangement.

Advanced Case Conference

The referral. A 26-year-old postgraduate student is referred by her general practitioner with "anxiety and low mood". She reports six months of difficulty sleeping, poor concentration and reluctance to attend seminars.

The assessment. Interview establishes that the difficulty began after a panic attack in a lecture theatre eight months earlier. She now avoids lecture theatres, travels by a longer route that avoids the underground, and has stopped attending seminars. She carries water at all times and sits near exits. She has not had a second full panic attack, but experiences near-attacks several times a week. She describes low mood and anhedonia over the last three months, and scores in the moderately severe range on a depression measure and the moderate range on an anxiety measure. She reports no elevated mood, no obsessions or compulsions, and no trauma history. She drinks alcohol only socially.

The reasoning. The presentation is best explained by panic disorder with agoraphobic avoidance, with a secondary depressive episode that developed as the avoidance restricted her life. The reverse order — primary depression with secondary anxiety — is less consistent with the clear onset at a panic attack and the specific situational avoidance. The near-attacks and safety behaviours indicate that the panic disorder remains active despite the absence of full attacks. The depression is clinically significant and would need treatment in its own right if it persisted after the panic was addressed.

The decision. The report records panic disorder with agoraphobia and a moderate depressive episode, identifies the avoidance and safety behaviours as the principal maintaining processes, and recommends a course of cognitive behavioural therapy with interoceptive and situational exposure, alongside monitoring of mood. It recommends review at eight weeks, with a decision about separate treatment for depression if mood has not improved as the avoidance reduces.

The teaching point. The presenting complaint was "anxiety and low mood"; the assessment found a specific, treatable fear structure. The functional analysis of avoidance generated the treatment plan, and the diagnostic sequence determined which condition to treat first.

Chapter Summary

Assessment of the internalising disorders requires specific differential reasoning rather than general enquiry. Depression must be assessed for course, subtype, severity and bipolarity, and bipolar disorder must be actively excluded. Anxiety presentations are distinguished by their fear structure, which also determines the treatment target. Obsessive-compulsive disorder is defined by obsessions and by rituals that are frequently covert. Trauma exposure is not equivalent to post-traumatic stress disorder, and assessment must establish both. Adjustment disorder is defined by exclusion and is vulnerable to over-use. Severity measures support monitoring but do not establish diagnosis. The functional analysis of avoidance links assessment to treatment, and the integrated formulation should state maintaining mechanisms in a form that predicts response.

Review Questions

1. Why is a severity score insufficient to establish the diagnosis of a depressive episode?
2. Distinguish panic disorder from generalised anxiety disorder in terms of the fear structure.
3. What are covert compulsions, and why are they important to ask about?
4. Explain why trauma exposure does not establish post-traumatic stress disorder.
5. What is the difference between capacity and performance in a functional analysis of avoidance?

Discussion Questions

1. A client's depression score is in the mild range, but she has stopped working and is drinking heavily. How do you report severity?
2. How should an assessor respond when a service requires a single diagnosis for a client who meets criteria for three?
3. Is adjustment disorder a useful diagnosis? Argue both sides.

4. How would you assess a client whose panic symptoms might be cardiac in origin?
5. What are the risks of using a cut score to determine access to treatment?

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- Placeholder reference entry.

GLOSSARY

Glossary

Term. Placeholder definition.

APPENDIX

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